

Glutamatergic Supramammillary Nucleus Neurons Respond to Threatening Stressors and Promote Active Coping

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Abstract

Threat-response neural circuits are conserved across species and play roles in normal behavior and psychiatric diseases. Maladaptive changes in these neural circuits contribute to stress, mood, and anxiety disorders. Active coping in response to stressors is a psychosocial factor associated with resilience against stress induced mood and anxiety disorders. The neural circuitry underlying active coping is poorly understood, but the functioning of these circuits could be key for overcoming anxiety and related disorders. The supramammillary nucleus (SuM) has been suggested to be engaged by threat. SuM has many projections and contains poorly understood diversity of populations. We identified a unique population of glutamatergic SuM neurons (SuM^{VGLUT2+}::POA) based on projection to the POA and found SuM^{VGLUT2+}::POA neurons have extensive arborizations. SuM^{VGLUT2+}::POA neurons project to brain areas that mediate various features of the stress and threat responses including the paraventricular nucleus thalamus (PVT), periaqueductal gray (PAG), and the habenula (Hb). Thus, SuM^{VGLUT2+}::POA neurons are positioned as hubs connecting to areas implicated regulating stress responses. Here we report SuM^{VGLUT2+}::POA neurons are recruited by diverse threatening stressors and recruitment of SuM^{VGLUT2+}::POA neurons correlated with active coping behaviors. We found that selective photoactivation of the SuM^{VGLUT2+}::POA population drove aversion but not anxiety like behaviors. Activation of SuM^{VGLUT2+}::POA neurons in the absence of acute stressors evoked active coping behaviors and drove goal directed behavior. Also, activation of SuM^{VGLUT2+}::POA neurons was sufficient to convert passive coping strategies to active behaviors during acute stress. These findings establish a new node in stress response circuitry that has projections to many brain areas and evokes flexible active coping behaviors and offers new opportunities for furthering our neurobiological understanding of stress.

eLife assessment

This **valuable** manuscript investigates the role of a subpopulation of glutamatergic neurons in the supramammillary nucleus that projects to the pre-optic hypothalamus area in active coping but not locomotor activity. It provides **solid** evidence from experiments using fibre photometry or photostimulation during threatening tasks that these neurons allow animals to produce flexible behaviours in response to stress. However, the evidence is **incomplete** in several ways, including validation and quantification of anatomical tracing data that serve as a basis for the behavioral testing, the use of statistics, sex as a biological variable, genotype differences between experimental and control groups in behavioral tests, limiting its broad interest to neuroscientists across sub-fields.

Introduction

Threat-response neural circuits are conserved across species and have roles in normal behaviors and psychiatric diseases(1-4). Identifying and responding to threatening stressor is critical for survival but maladaptive changes in underlying neural circuits contribute to stress, mood, and anxiety disorders (5, 6). Active coping in response to stressors is psychosocial factor associated with resilience against stress induced mood and anxiety disorders(7). Available evidence indicates that active (e.g., escape, fighting) and passive (e.g., freezing, immobility) coping responses to stressors are governed by separable neural circuits (1, 8, 9). The functioning of circuits underlying active coping could be key for overcoming anxiety and related disorders(10, 11). The neural circuits, cells, and mechanisms underlying active coping strategies remain unclear(12, 13).

The supramammillary nucleus (SuM) has been suggested to be engaged by threatening stressors and has efferent connections to stress-sensitive brain regions so may be an important regulator of responses to stressors (14-16). Research on SuM has focused on connections to the hippocampus and septum while SuM projections to other brain areas remain less understood(15, 17, 18). SuM contains distinct populations with functionally diverse roles including regulation of hippocampal activity during REM, spatial memory, arousal, and environmental interactions(19-23). In addition to functional diversity, anatomical and molecular-cellular diversity is present in SuM(24-26). Divergent populations in SuM have been defined by differential to the dentate gyrus and CA2 regions of the hippocampus, and neurochemically based on neurotransmitter expression (27, 28). Major projections from the SuM, which have yet to be examined, include the preoptic hypothalamus area (POA). We examined if SuM neurons projected to the POA and if this projection could be used to aid in separation of populations in SuM.

We used retrograde adeno associated viral (AAV) and combinatorial genetic tools to identify and characterize a population of glutamatergic SuM neurons with projections to the POA (SuM^{VGLUT2+}::POA). We found that SuM^{VGLUT2+}::POA neurons represented an anatomical subset of SuM neurons with extensive arborizations to brain regions including that mediate stress and threat responses including the paraventricular nucleus (PVT), periaqueductal gray (PAG), and the habenula (Hb). Thus, SuM^{VGLUT2+}::POA neurons are positioned as hubs with spokes to many areas regulating responses to threatening stressors. We hypothesized this population could respond to and regulate responses to stressors. We found SuM^{VGLUT2+}::POA neurons are recruited by multiple types of acute threatening stressors, encode a negative valence, do not promote anxiety-like behaviors, and evoke active coping behaviors. Further, activation of these SuM^{VGLUT2+}::POA

neurons was sufficient to convert passive coping strategies to active behaviors. These findings indicate SuM^{VGLUT2+}::POA neurons are a central hub linked to multiple stress and threat responsive areas and can drive state transitions between passive and active responses to stress.

Results

VGLUT2+ SuM neurons projecting to the POA (SuM^{VGLUT2+}::POA) arborize to multiple stress-engaged brain regions

The SuM contains functionally diverse and anatomically distinct populations with efferent projections to many brain regions(16, 24, 38, 39). The diversity includes glutamatergic, GABAergic, and co-expressing GABAergic/Glutamatergic populations(21, 40). Using retrograde adeno associated virus (Retro-AAV) (Retro-AAV2-DIO-tdTomato) and anterograde (AAV5-DIO-ChR2eYFP) tracing we identified a population of VGLUT2+ expressing neurons in SuM that project to the POA (SuM^{VGLUT2+}::POA) with dense projections in the lateral preoptic area (LPO) within the POA (**Supplemental Figure 1A-F**) in VGLUT2-Cre mice (n=4). Using a viral construct encoding for a nuclear restricted fluorophore that switches from mCherry (red) to eGFP (green) in a Cre- dependent manner (Retro-AAV2-Nuc-flox(mCherry)-eGFP) we found SuM^{VGLUT2+}::POA neurons were positive for VGLUT2+ in VGLUT2-Cre mice (n=4) and negative for VGAT expression in VGAT-Cre mice (n=3) (**Supplemental Figure 1G-L**) indicating that SuM^{VGLUT2+}::POA neurons do not belong to a GABAergic/Glutamatergic populations.

To examine arborization of SuM^{VGLUT2+}::POA neurons we utilized a combinatorial genetic approach. Specifically, we used mice (VGLUT2-Flp) that express Flp recombinase in VGLUT2 expressing cells in a combination with Flp dependent expression (fDIO) of Cre and Cre (DIO) dependent fluorophore expression. We injected Retro-AAV2-DIO-eYFP into POA and AAV-fDIO-Cre in to SuM (**Figure 1A**). We thus expressed Cre in SuM^{VGLUT2+} neurons and, of those, only neurons projecting to the POA (SuM^{VGLUT2+}::POA neurons) were labeled with eYFP. We found SuM^{VGLUT2+}::POA neurons arborize widely projection to multiple brain regions including: the nucleus accumbens (Acb), Septum, lateral hypothalamus (LH), ventral medial hypothalamus (VMH), paraventricular nucleus (PVT), lateral habenula (LHb), the ventral lateral periaqueductal gray (VLPAG), dorsal raphe (DRD), lateral parabrachial nucleus (LPBN), laterodorsal tegmental nucleus (LDTg), and medial vestibular nucleus (MVe) (**Figure 1 D-M**). We visualized arborization of SuM^{VGLUT2+}::POA neurons in cleared tissue by using optical light sheet imaging. For these experiments we injected VGLUT2-Flp mice (n=3) we injected as in **Figure 1A** and brains were actively cleared using SHIELD for optical light sheet imaging(42). Compiled three-dimensional images of a brain hemisphere, viewed from the medial to lateral perspective (**Figure 1M**) or viewed from the ventral to dorsal perspective (**Supplemental Figure 2M**), showed projections labeled by eYFP from SuM^{VGLUT2+}::POA to POA, areas of hippocampus, septum, Acb, and regions in the pons and midbrain. The results further established projections from SuM^{VGLUT2+}::POA neurons to multiple brains areas and illustrate the broad arborization.

We used tracing with Retro-AAV's to confirm our findings from anterograde tracing by injecting unilaterally into the Acb, septum, PVT, or PAG in VGLUT2-Cre mice. Cell bodies in SuM were labeled eYFP or tdTomato (**Supplemental Figure 2A-L**). Each area was injected in three or more mice yielding similar results. We examine anatomic distribution of SuM::POA neuron in SuM by injected Retro-AAV2-Cre into the POA and Retro-AAV2-Nuc-flox(mCherry)-eGFP into the SuM (**Supplemental Figure 3 A-B**) of wildtype (WT) Cre- mice (n=3) injected with this combination of viruses we observed both neurons labeled by mCherry (Cre negative) or eGFP (Cre expressing) interspersed in the SuM. We verified the combinatorial viral selectively labeled SuM^{VGLUT2+}::POA neurons with minimal background(43). Positive controls we injected Retro-AAV-Flpo into the POA and AAV-fDIO-Cre into the SuM labeling of cells in SuM (**Supplement Figure 3 G and H**). As

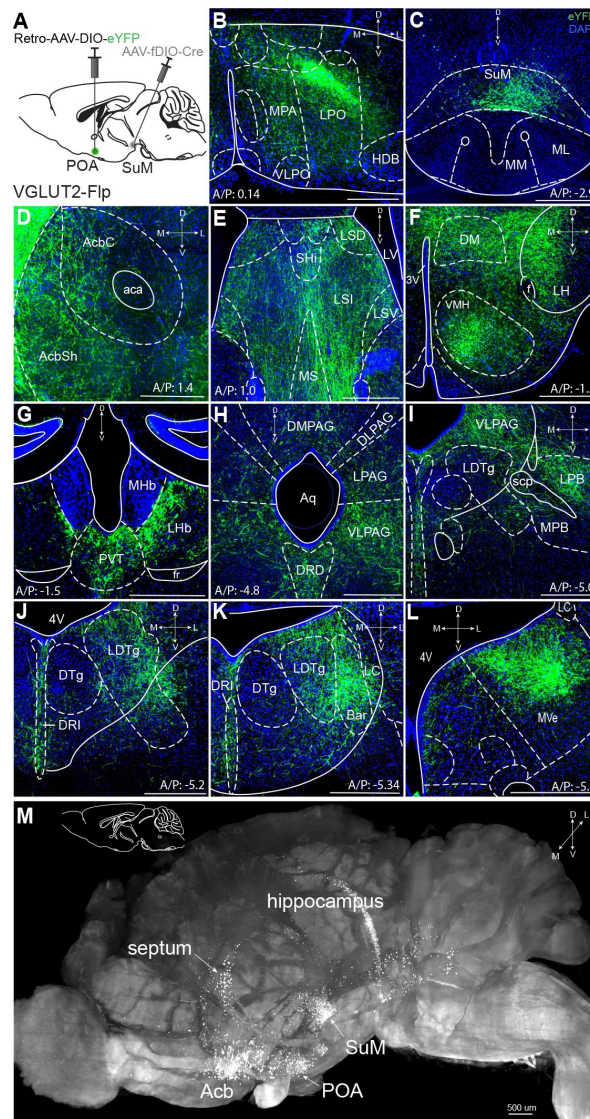


Figure 1.

SuM^{VGLUT2+}::POA arborize widely in the brain.

(A) Schematic of injections of Retro-AAV-DIO-eYFP in to the POA and AAV-fDIO-Cre into the SuM of VGLUT2-Flp mice to label only VGLUT⁺ SuM neurons that project to the POA. (B) Projections of VGLUT⁺ SuM neurons labeled with eYFP seen in the POA. (C) Cell bodies labeled with eYFP in the SuM. (D-L) Arborizing processes from the SuM^{VGLUT2+}::POA neurons are seen in multiple brain regions including the (D) AcbSh and AcbC; (E) lateral septum; (F) multiple hypothalamic areas; (G) LHb and PVT; (H) the PAG and the DRD; (I) VLPAG and LPB; (J) DRI and LDTg; (K) and DRI, and MVePC. (M) Light sheet microscopy image of a cleared mouse brain hemisphere, viewed from medial to lateral, showing eYFP labeled neurons in SuM^{VGLUT2+}::POA neurons with cell bodies in SuM and processes in areas corresponding to septum, hippocampus, Acb, and POA. (500 μ m scale bars)

Abbreviations: MPA- medial preoptic, VLPO- ventral lateral preoptic, LPO- lateral preoptic, HDB- nucleus of the horizontal limb of the diagonal band, AcbSh- Accumbens shell, AcbC- Accumbens core, Shi- septohippocampal nucleus, LSI- lateral septal nucleus, intermediate part, LSV- lateral septal nucleus, ventral part, LV- lateral ventricle, MS- medial septal nucleus, LHb- lateral habenula, MHb- medial habenula, PVT- paraventricular thalamus, DM- dorsomedial hypothalamic nucleus, LH- lateral hypothalamic area, VMH- ventromedial hypothalamic nucleus, PAG- periaqueductal gray, DMPAG- dorsomedial periaqueductal gray, DLPAG- dorsolateral periaqueductal gray, LDTg- laterodorsal tegmental nucleus, LPAG- lateral periaqueductal gray, DRD- dorsal raphe nucleus, dorsal part, LPB- lateral parabrachial nucleus, MPB- medial parabrachial, scp- superior cerebellar peduncle, Bar- Barrington's nucleus, DTg- dorsal tegmental nucleus, DRI- dorsal raphe, interfascicular part, LC- locus coeruleus, MVe- medial vestibular nucleus, 4V- fourth ventricle

negative controls, we injected only AVV-fDIO-Cre into the SuM in Ai14 mice, which did not yield tdTomato expression (**Supplemental Figure 3C-D**). To confirm that the combination of Retro-AAV2-fDIO-Cres and AAV5-DIO-ChR2eYFP did not lead to labeling of cells with eYFP in the absence of Flp we injected Retro-AAV2-fDIO-Cre into the POA and AAV-DIOChR2eYFP into SuM of WT mice. In these mice we did not observe any expression of eYFP (**Supplemental Figure 3E-F**). The results demonstrate the specificity of our combinatorial viral strategy. All studies replicated in a minimum of three mice.

SuM^{VGLUT2+}::POA neurons are an anatomical distinct subset of all SuM^{VGLUT2+} neurons

Recent studies have highlighted functionally divergent roles of SuM neurons and suggested differential projections, particularly to regions of hippocampus and PVT, may identify functionally distinct populations ([19](#), [20](#)). To examine if SuM^{VGLUT2+}::POA neurons are a subset of SuM^{VGLUT2+} neurons we qualitatively compared the projections of total SuM^{VGLUT2+} neurons to SuM^{VGLUT2+}::POA neurons. Similar to projections for SuM^{VGLUT2+}::POA neurons (**Figure 1**) we observed labeled projections from the total SuM^{VGLUT2+} neurons in the POA, PVT, LHB, and the CA2 field of the hippocampus (**Figure 2 A-E**). Importantly, we found areas that received projections from the total SuM^{VGLUT2+} and not from SuM^{VGLUT2+}::POA neurons. As schematized (**Figure 2K**), projections from the total SuM^{VGLUT2+} but not SuM^{VGLUT2+}::POA populations were present in dentate gyrus (DG) and medial habenula (mHb) (**Figure 1G** and **Figure 2 F-J**). Thus, SuM^{VGLUT2+}::POA population can be further defined as a subset of the total SuM^{VGLUT2+} neuronal population with distinct projection targets. Some areas, such as septum, may receive projections from multiple populations.

Threatening stressors not spontaneous higher velocity movement recruit SuM^{VGLUT2+}::POA neurons

Neurons in SuM can be activated by acute stressors, and we sought examine the recruitment of SuM^{VGLUT2+}::POA neurons by threatening stressors ([14](#), [44](#), [45](#)). We found that forced swimming induced cFos expression in SuM. Labeling SuM^{VGLUT2+}::POA neurons using Retro-AAV-DIO-mCherry revealed an increase in the number of mCherry and cFos labeled cells following forced swim (**Supplemental Figure 4A-G**). To further examine recruitment of SuM^{VGLUT2+}::POA neurons by acute stressors we tested a diverse set of acute threatening stressors using *in vivo* Ca²⁺ detection via fiber photometry. We expressed GCaMP7s in SuM^{VGLUT2+}::POA neurons (**Figure 3A-B**) using the combinatorial viral genetic approach detailed for anatomic studies. We developed a forced swim assay that utilized a moveable platform allowing mice to be placed into and removed from water while obtaining fiber photometry recordings. Mice were dunked for a 30-second swim trial every 2 minutes for a total of 10 trials (n=8 mice). During the assay mice exhibited almost exclusively active swimming and climbing behaviors reflected in the quantification of mean time mobile (**Supplemental Figure 4I**). Analysis of Ca²⁺ dependent (470 nm excitation) and isosbestic control (415 nm excitation) signals from fiber photometry recordings reveal rapid rise of approximately 5 standard deviations in GCaMP Ca²⁺ dependent signal with start of the swim trial that was sustained through the 30-sec swim period. The Ca²⁺ dependent signal returned to baseline following the platform raising removing the animal from the water (**Figure 3F and G**). These data show the dunk assay evoked active coping behavior and recruited SuM^{VGLUT2+}::POA neurons.

We also tested a foot shock paradigm (**Figure 3D**) and a predator ambush assay (**Figure 3E**). In foot shock assays VGLUT2-Flp mice (n=6), prepared as above (**Figure 3A and B**), were subjected to five pseudo randomly spaced trials with 30 sec of tone proceeding a two-second shock. We observed a rapid increase in the Ca²⁺ dependent GCaMP7s signal following the foot shock (**Figure 3H and I**). For the predator ambush we adapted a previously demonstrated

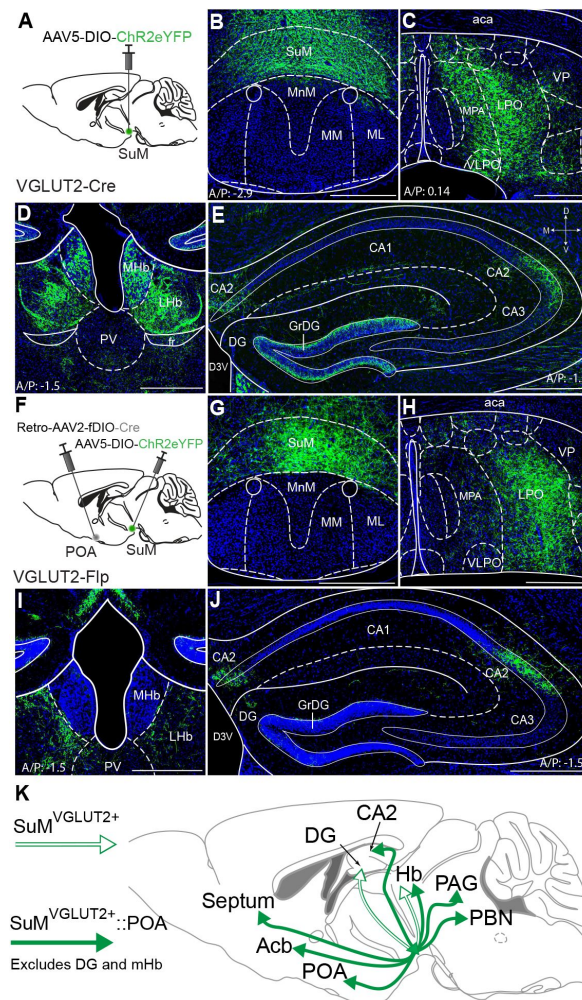


Figure 2.

SuM^{VGLUT2+}::POA neurons project to a subset of brain regions compared to all SuM^{VGLUT2+} neurons.

(A) Schematic of injections in VGLUT2-Cre mice of AAV5-DIO-ChR2eYFP into SuM. (B) eYFP (green) labeled neurons present in SuM and (C) processes were observed in the POA. (D) Process from SuM VGLUT2+ neurons were also present in MHb, LHb and PVT. (E) In hippocampus process were observed in DG and CA2. (F) Schematic of injections in VGLUT2-Flp mice of Retro-AAV2-fDIO-Cre and AAV5-DIO-ChR2eYFP into SuM. (G-H) Cells in SuM and process in POA were labeled by ChR2eYFP. (I) Labeled processes in LHb were evident but none seen in MHb. (J) In hippocampus, labeled processes were present in CA2 but not observed in DG. (500 μ m scale bars) (K) Schematic summary showing the projection to DG and MHb present in the total SuM VGLUT2+ population (outlined arrows) but absent in the SuM^{VGLUT2+}::POA population (filled arrows).

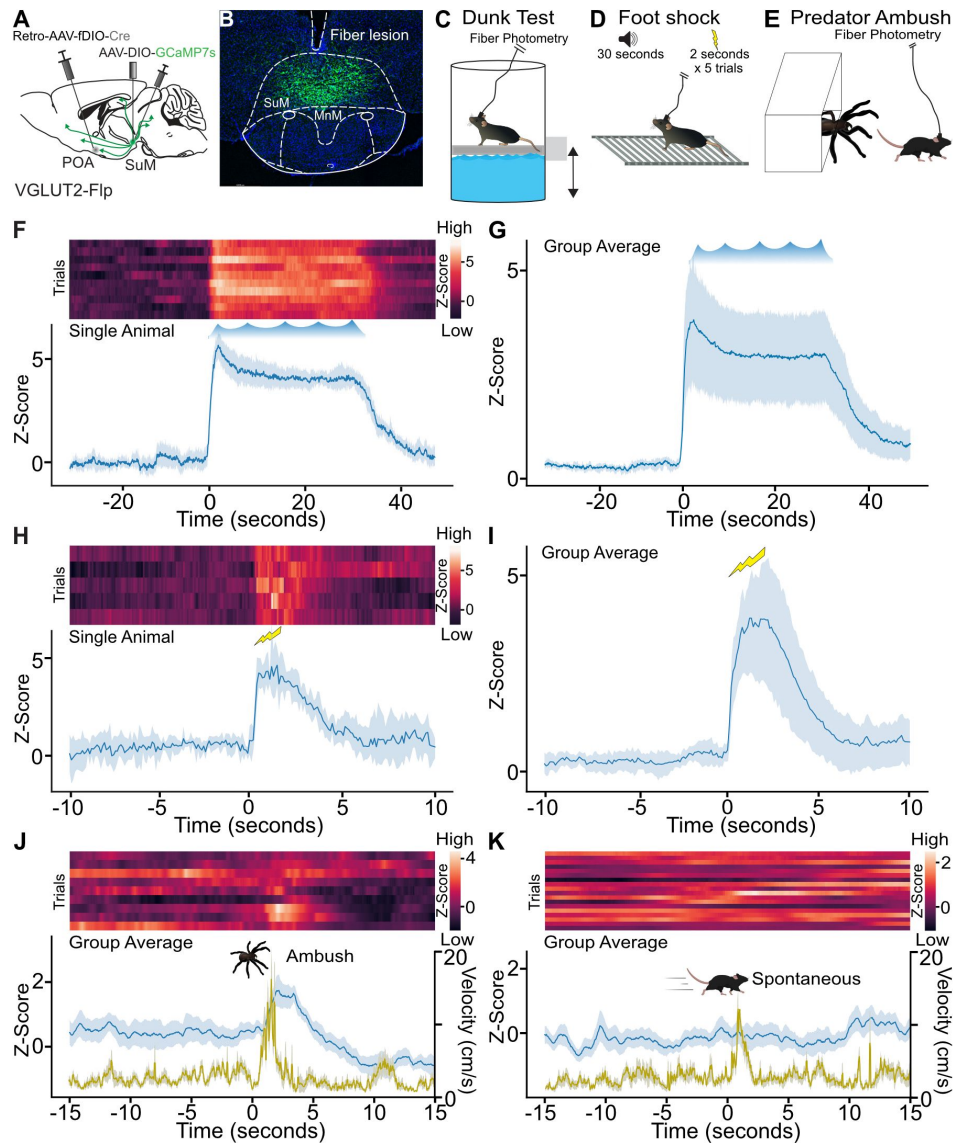


Figure 3.

Acute stressors recruit SuM^{VGLUT2+::POA} neurons.

(A) Schematic of VGLUT2-Flp mice injected with AAV-DIO-GCaMP7s in SuM and Retro-AAV-fDIO-Cre in the POA with a (B) fiber cec over SuM for fiber photometry recordings from SuM^{VGLUT2+::POA} neurons. (C) While mice were connected for fiber photometry they were (C) subjected to ten 30-second trials of forced swimming, (D) to a 2-second foot shock following a 30 second tone for five trials, or (E) to mock predator ambushed by remote-controlled spider. (F) Heat map and mean $\pm$ 95% CI Z-score for recordings obtained from single animal during the repeated forced swim showing increase Ca^{2+} signal during the swim session. (G) The mean $\pm$ 95% CI Z-score of 10 trials for all animals ($n=8$) in the dunk assay. (H) Heat map and an $\pm$ 95% CI Z-score for recordings from a single animal during the five shock trials showing increase Ca^{2+} signal. (I) Mean $\pm$ 95% CI Z-score for recordings of five trials for all animals ($n=6$) in the foot shock assay. (J) Heat map, mean $\pm$ 95% CI Z-score (blue), mean $\pm$ 95% CI velocity (gold), for recordings obtained from animals ($n=9$) during the ambush showing an increase in Ca^{2+} signal as the animals flee from the remote-controlled spider with the mean $\pm$ 95% confidence interval Z-score of 1 trial bush) for all 9 animals. (K) Heat map, mean ($\pm$ 95%CI) Z-score (blue), mean ($\pm$ 95% CI) velocity (gold), for the same mice but ambush in the predator assay. Time frame gated for spontaneous locomotion. Ca^{2+} signal does not increase during spontaneous locomotion. Mean peak velocity was not significantly different.

paradigm using a mock mechanical spider attack (46). In this assay a remote-controlled mechanical spider was hidden in a box with swing door. The box was inside a larger walled arena. Mice were able to freely explore the arena and at a moment when the mouse was in proximity to the box opening the spider was moved out towards the mouse. The mice fled, often to a corner, stopped, turned to face the spider (**Supplemental video 1**). In mice (n=9) subjected to the ambush paradigm we observed an increase in Ca^{2+} dependent GCaMP signal at the time of the ambush followed by suppression below the initial baseline (**Figure 3J**). Populations of neurons in SuM have previously been found to correlate with future movement velocity (41), so we examined if recruitment of SuM^{VGLUT2+}::POA neurons was correlated to periods of spontaneous higher velocity movement. Analyzing data collected prior to the ambush event we examined if increased velocity, at speeds similar to the fleeing induced by the ambush, was correlated with increased Ca^{2+} dependent GCaMP signal in SuM^{VGLUT2+}::POA neurons (**Figure 3L**). We found no correlation (**Figure 3K**). We further examined the data for a correlation of increased Ca^{2+} dependent signal in SuM^{VGLUT2+}::POA neurons to movement velocity during open field exploration in a sperate cohort of mice (n=13). Here we found no evidence for correlation of velocity with Z-score of Ca^{2+} dependent signal including a cross correlation analysis to account for a potential temporal offset (**Supplemental Figure 5**). In aggregate, the data support recruitment of SuM^{VGLUT2+}::POA neurons by diverse threatening stressors but not during times of higher velocity spontaneous movement.

SuM^{VGLUT2+}::POA neurons evoke active coping behaviors

To examine how activation of SuM^{VGLUT2+}::POA neurons contributes to responding to threatening stressors we assessed behavioral changes evoked by photostimulation of SuM^{VGLUT2+}::POA neurons. We injected VGLUT2-Cre or WT (Cre-) mice with Retro-AAV5-DIO-ChR2eYFP in the POA, and an optic fiber placed over SuM (**Figure 4A**). We employed a paradigm with a 15-minute trial divided into three 5-minute periods, pre-stimulation, stimulation at 10 Hz, and post-stimulation. Based on review of videos obtained during the 15-min trials, we observed behaviors that could be classified into nine distinct categories: grooming, immobile, walking, chewing of bedding, rearing, rapid locomotion (movement was limited by the size of the arena), digging (moving bedding towards the tail), treading (moving bedding forward with front paws), and jumping (**Figure 4B**). A blinded observer scored the behaviors in 10-second intervals for the predominant behavior observed during each interval and each behavior was assigned a color for visualization. Color coded representation of trials from 16 Cre+ (n=16) and WT (n=16) mice is illustrated (**Figure 4B**). The behavioral pattern in the pre-stimulation period is similar between the Cre+ and WT mice. During the stimulation period, a clear shift in behavior was evident in Cre+ mice. Photostimulation induced rearing, treading, digging, rapid locomotion, and jumping. During the post-stimulation period we observed a new pattern with Cre+ mice spending time immobile and grooming. An example of a Cre+ animal is shown in **Supplemental video 2**.

We analyzed behavior quantifying specific behaviors during each period. Quantification of jumping revealed that Cre+ mice (n=16) engaged in significantly (p=0.0012) more jumps during the stimulation period compared to WT (n=16) mice (**Figure 5C**). During the pre- and post-stimulation periods there was not a significant difference in the number of jumps. We also observed that during the stimulation period the Cre+ mice engaged in bouts treading/digging, vigorously moving bedding forwards and backwards, as previously described for defensive burying (47). Defensive burying, characterized by moving bedding forward or backwards often in alternating pattern, is evoked by threatening and noxious stimuli, and is a described active stress coping strategy in rodents (47-50). We conservatively quantified together the movement of bedding as treading/digging that may include spontaneous digging (**Figure 4D**) and found no significant (p=0.7) difference during the pre-stimulation period. In the stimulation period there was a significant (p=0.004) increase in the time spent treading/digging, and surprisingly, during the post-stimulation period time treading/digging was significantly (p=0.0028) decreased in Cre+ mice. The variability in behaviors can be attributed to exclusion of one behavior

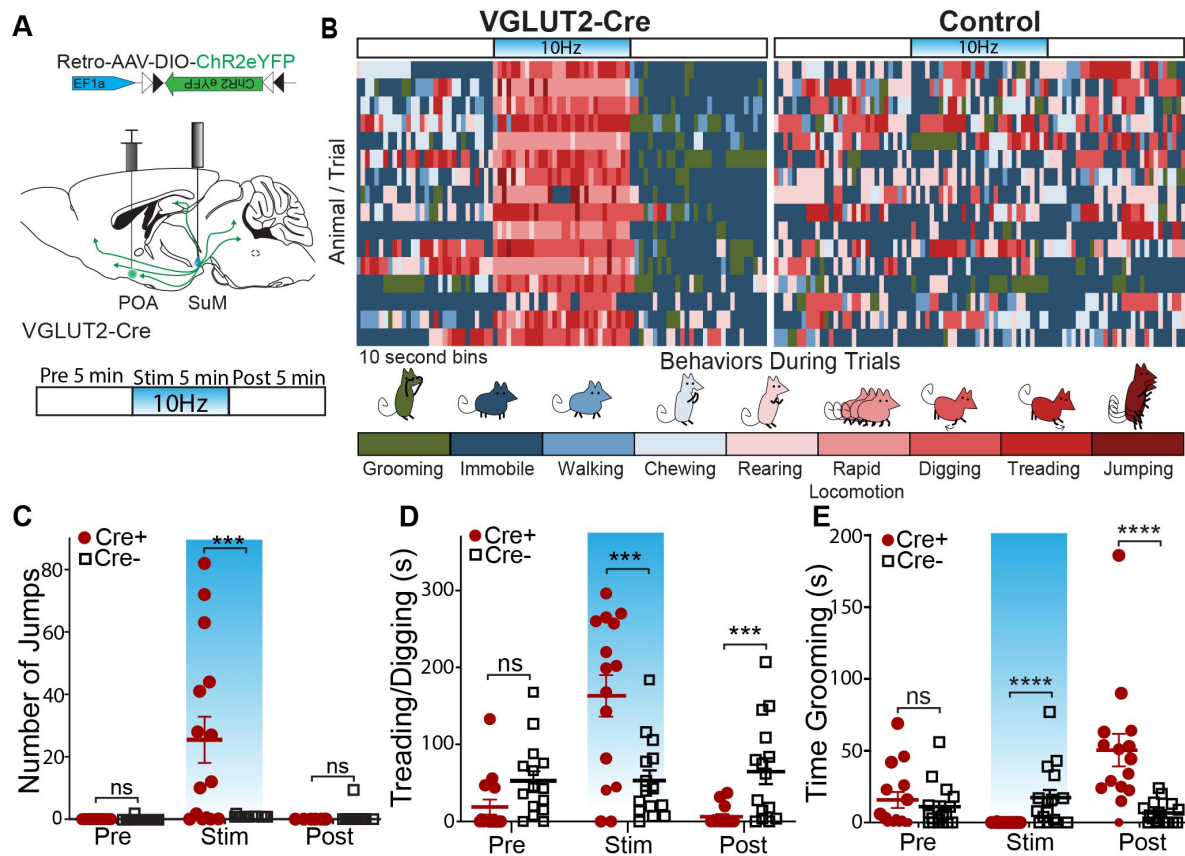


Figure 4.

Photostimulation of SuM^{VGLUT2+::POA} neurons evokes active coping behaviors.

(A) Illustration of injections and fiber implant in VGLUT2+ Cre mice and schematic of photostimulation paradigm of 5 min pre, 5 min stimulation, and 5 mins post. Behaviors was evaluated during each epoch. (B) For Cre+ (n=16) and Cre- (n=16) mice the behavior in 10 seconds bins was scored based on the predominate behavior displayed during each 10 period for the 15-minute trial into grooming, stationary, walking, chewing, rearing, rapid locomotion, digging (moving bedding towards the tail), treading (move beading forward), and jumping. The graphic show a by animal scoring of the 15-minute trial color coded for each of the behavior categories. (C) During the stimulation period Cre+ (n=15) mice show significantly ($p=0.0012$) greater jumps than Cre- (n=16) mice and jumping behaviors was not significantly different during pre and post stim periods. (D) Behavior was also scored for time spent moving beading (digging or treading). Cre+ mice showed significantly ($p=0.004$) greater time engaging in digging/ treading behaviors during the stimulation period and significantly ($p=0.002$) less time during the post stimulation period compared to Cre- mice. (E) Behavior was also scored for time spent engaging in grooming behaviors and Cre+ mice spent significantly ($p<0.001$) less time grooming during the stimulation period and significantly ($p<0.001$) more during the post stimulation period compared to Cre- mice. All data plotted as mean $\pm$ SEM

by the other, with individual mice having variability in the predominant behavior displayed during the photostimulation period but all mice shifted to increased activity coping behaviors. The behaviors evoked by photostimulation fit with active coping behaviors. Specifically, escape (jumping, rapid locomotion), defensive burying/treading (digging, pushing beading forward).

We also quantified grooming behaviors in each of the three periods (**Figure 4E**). During the pre-stimulation period there was not a significant difference. During the stimulation period the Cre⁺ mice did not engage in grooming leading to a significant ($p < 0.0001$) decrease in time spent grooming compared to Cre⁻ mice. In the post-stimulation-period Cre⁺ mice showed a significant ($p < 0.0001$) increase in time spent grooming compared to Cre⁻ mice. A reasonable interpretation of rise in grooming post stimulation is that photostimulation evoked a stressed like state and cessation of photostimulation led to selfcare grooming as seen following acute stressors (51, 52). In summary, the analysis of behaviors elicited by photostimulation of SuM^{VGLUT2⁺}::POA neurons without conditioned cues or concomitant stressors demonstrates a dramatic shift in behavior to escape oriented (jumping, rapid locomotion) and threat response behavior, including rearing and defensive burying, during the photostimulation period. In contrast to freezing, the behaviors elicited by activation of SuM^{VGLUT2⁺}::POA neurons indicate that they alter active coping strategies.

Photostimulation of SuM^{VGLUT2⁺}::POA neurons drives real-time avoidance

To examine whether activation of SuM^{VGLUT2⁺}::POA neurons may contribute to the aversive aspects of threatening stress we carried out real-time place aversion testing (RTPA) by pairing one side of the chamber with photostimulation of SuM^{VGLUT2⁺}::POA (**Figure 5A**) neurons at multiple frequencies (1, 5, 10, and 20 Hz). Photostimulation of SuM^{VGLUT2⁺}::POA neurons produced significant ($p < 0.0001$) aversion at all frequencies in Cre⁺ ($n = 19$) mice compared to Cre⁻ ($n = 21$) control and higher stimulation frequencies evoked greater aversion (**Figure 5B and C**). Example of Cre⁺ mice with stimulation at 1, 5, and 10 Hz as well as Cre⁻ mice is in **Supplemental Video 3**. WT and Cre⁺ mice explored equivalently, with similar ($p = 0.87$ mixed-effects model) number of entries to the stim side (**Figure 5D**). Cre⁺ mice quickly left the stimulation side and the mean time spent on the stimulation side of the area was significantly ($p < 0.0001$) lower (**Figure 5E**). These data indicate surprisingly that photostimulation, although aversive, did not generate aversive pairing, as found for other brain areas (53). To examine the relative aversiveness of SuM^{VGLUT2⁺}::POA photostimulation we carried out RTPA experiments using an arena with a dark side and a bright side (**Figure 5F inset**). We paired photostimulation with the dark side. As expected, Cre⁻ mice show a preference for the dark side of the arena (**Figure 5F**), however photostimulation of the SuM^{VGLUT2⁺}::POA neurons yielded significant (1 Hz $p = 0.002$, 5 Hz $p = 0.0004$, and 10, 20 Hz $p < 0.0001$) aversion of Cre⁺ ($n = 13$) compared to Cre⁻ ($n = 12$). Cre⁺ mice spent nearly the entire trial in the brightly light side demonstrating photostimulation of SuM^{VGLUT2⁺}::POA neurons drove avoidance sufficient overcome a mildly aversive stimulus (bright light).

Reports have described multiple populations of neurons in supramammillary including neuronal populations that project to the dentate gyrus of the hippocampus that release both GABA and glutamate. Our anatomic studies indicated that SuM^{VGLUT2⁺}::POA neurons do not project to the dentate gyrus (**Figure 2**) and are not GABAergic (**Supplemental Figure 1G-L**). Thus, SuM^{VGLUT2⁺}::POA neurons are not part of either a GABAergic or dual transmitter population. To examine if GABAergic neurons in SuM (SuM^{VGAT⁺}) can mediate real-time place aversion or preference, we injected VGAT-Cre mice with AAV to express ChR2 ($n = 26$) or control ($n = 26$) in SuM and carried out real-time place aversion testing. Photostimulation at 10 Hz yielded no significant aversion or preference compared to baseline (**Supplemental Figure 6A-D**). This is in contrast to the robust aversion caused by photostimulation of SuM^{VGLUT2⁺}::POA neurons.

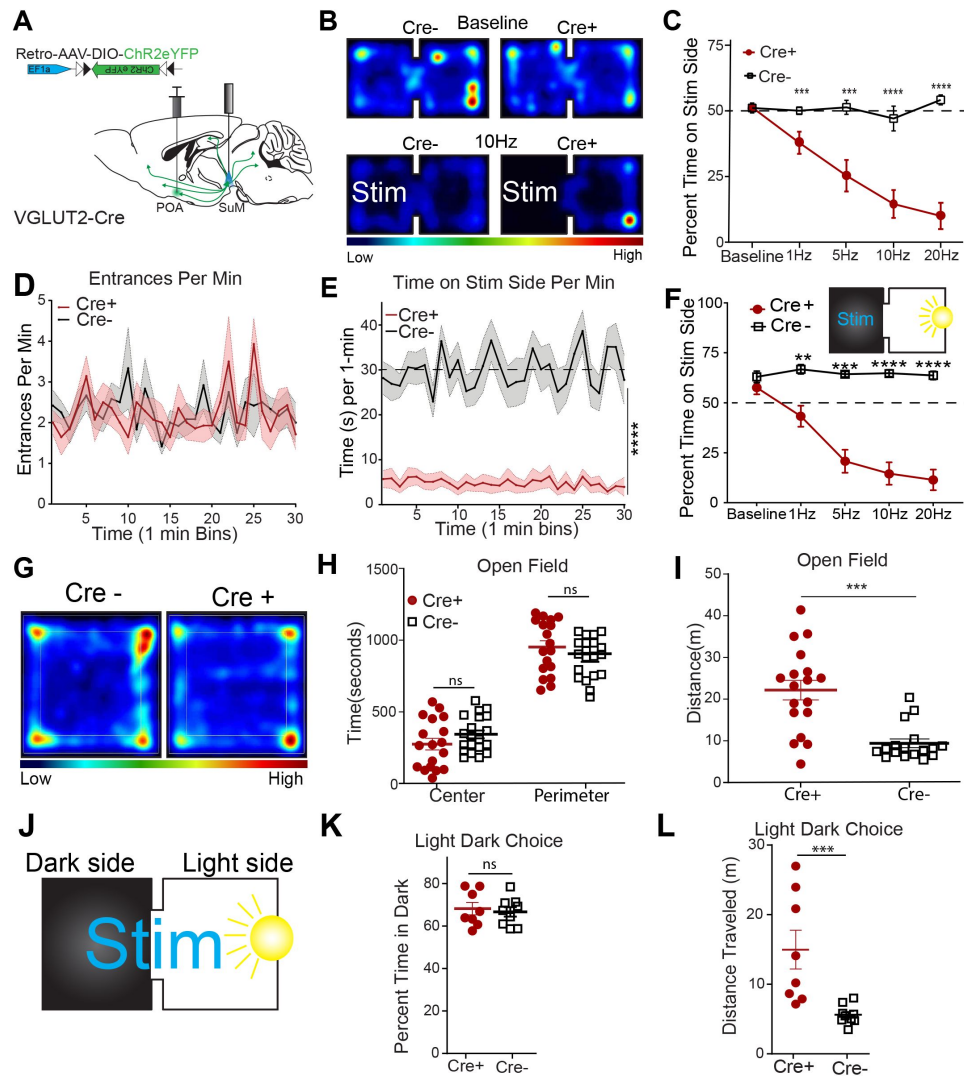


Figure 5.

Activation of SuM^{VGLUT2+}::POA neurons drives aversion but does not promote anxiety-like behavior

(A) Schematic of injection of Retro-AAV-ChR2eYFP into the POA and placement of a midline optic fiber over SuM. (B) Representative heat maps for Cre+ and Cre- mice at baseline (no stim) and 10Hz photostimulation in two-sided arena with photostimulation associated with one side. (C) Quantification of time spent of stimulation side for Cre+ (n=19) and control Cre- (n=21) mice showing frequency dependent increase in avoidance of the stimulation side of the area (***p<0.001, ****p<0.0001). (D) Mean $\pm$ SEM entrances per min (in 1-min bins) to the during 10 Hz stimulation trials were not significantly (p=0.87) different between Cre+ (n=14) and Cre- (n=13) mice. (E) Mean time spent on the stimulation side for Cre- and Cre+ in 1-minute bins during 10 Hz stimulation trials was significantly (p<0.0001) lower in Cre+ mice. (F) (Inset) Diagram of light/dark arena with photo stimulation provided on the dark side of the arena and quantification of time spent on dark (stimulation) side demonstrating that Cre- (n=12) animals show a baseline preference for the dark side of arena that is overcome by photostimulation with Cre+ (n=13) mice spending significantly (**p=0.002, ***p<0.001, ****p<0.0001) less time on the stimulation side. (G) Representative heat map of Cre- and Cre+ during 10Hz photostimulation in open field test. (H) In open field test time spent in center and perimeter were not significantly (p=0.19) between Cre+ (n=18) and Cre- (n=17) during 10 Hz photostimulation. (I) Distance traveled during open field testing was significantly (p<0.001) increased in Cre+ compared to Cre- mice. (J) Schematic of real time light/dark choice testing with stimulation provided at 10Hz throughout the arena. (K) Both Cre+ (n=8) and Cre- (n=9) mice showed a preference for the dark portion of the arena but were not significantly (p=0.9) different. (L) The total distance traveled by Cre+ mice were significantly (p=0.003) greater than Cre- mice. All data plotted as mean $\pm$ SEM.

Photoactivation of SuM^{VGLUT2+}::POA neurons does not cause anxiogenic-like behavior

Threat can induce anxiety-like, risk aversion behavioral states (54–56). To test whether activity of SuM^{VGLUT2+}::POA neurons contributes to anxiety-like behaviors we used two established assays for anxiety-like behavior, open field, and light-dark exploration (57–60). During open field testing 10 Hz photostimulation was applied to SuM^{VGLUT2+}::POA neurons. We quantified the time spent in the perimeter (outer 50%) vs. the center (inner 50%) of the arena and found VGLUT2-Cre⁺ (n=18) and WT (n=17) mice did not significantly (p=0.17, and 0.19) differ (Figure 5G and H). The total distance traveled was significantly (p<0.001) increased in Cre⁺ mice compared to WT mice (Figure 5I). Increased distance traveled in some mice was not surprising, because prior testing (Figure 4) revealed the mice engaged in escape behaviors during photostimulation. We observe behaviors including jumping during open field testing as well. We also carried out open field testing using VGAT-Cre mice to selectively activate SuM^{VGAT+} neurons and observed no significant (p=.99) effect on time in center or perimeter (Supplemental Figure 6F), but mice expressing ChR2eYFP displayed a small but significant (p=0.035) decrease in distance traveled (Supplemental Figure 6G). We tested if photostimulation of SuM^{VGLUT2+}::POA neurons would increase preference for the dark area, a potential sign of elevated anxiety (Figures 5J). We found photostimulation of SuM^{VGLUT2+}::POA neurons did not significantly (p=0.9) alter preference (Figure 5K) in VGLUT2-Cre⁺ mice (n=8) compared to WT (n=9). The activation of SuM^{VGLUT2+}::POA neurons evoked escape behaviors in open field testing, as seen in Figure 4, leading to a significant (p=0.003) increase in total distance moved (Figure 6L). The findings we obtained in open field and light-dark exploration testing do not support a role for SuM^{VGLUT2+}::POA neurons in driving anxiety-like behaviors.

SuM^{VGLUT2+}::POA neurons can drive instrumental action-outcome behavior

Current theoretical frameworks for examining threat responses divide behaviors into two broad categories, innate (fixed) and instrumental (action-outcome) (1, 61). Additionally, stress responses can be divided into active vs passive actions. Behaviors in these separable categories are mediated by distinct neural circuits (1, 62). For example, areas involved in responding to specific threatening stimuli (looming threat) drive an innate fixed behavior repertoire (10). Photostimulation of SuM^{VGLUT2+}::POA neurons evoked multiple active behaviors displayed when responded to threatening or noxious stimuli (Figure 4). To test if SuM^{VGLUT2+}::POA neurons can promote flexible repertoires of behavior including goal-directed, as opposed to only innate behaviors (ex. fleeing), we used an operant negative reinforcement paradigm (Figure 6A–B). Because our results show photostimulation of SuM^{VGLUT2+}::POA neurons is aversive photostimulation of SuM^{VGLUT2+}::POA neurons could be used as a negative reinforcer. We hypothesized if stimulation of SuM^{VGLUT2+}::POA evokes fixed innate defensive behaviors, or simply promotes high amounts of locomotion, then performance of an operant task (an action-outcome behavior) would be impaired because these behaviors would conflict with performance of the operant task. Importantly, results obtained can help differentiate general elevation in locomotor activity from directed coping behavior. We employed a paradigm using two nose poke ports, one active and one inactive. SuM^{VGLUT2+}::POA neurons were photostimulated at 10 Hz during a 10-minute trial. Activation of the active port by a nose poke triggered a 10-second pause in photostimulation and turned on a house light for 10 sec (Figure 6A and B). Following this 10-sec pause the house light switched off and photostimulation resumed until the subsequent activation of the active port. We tested animals for 4 consecutive days with the 10-min trials without prior training. Importantly, performance of the task had to be completed during photostimulation.

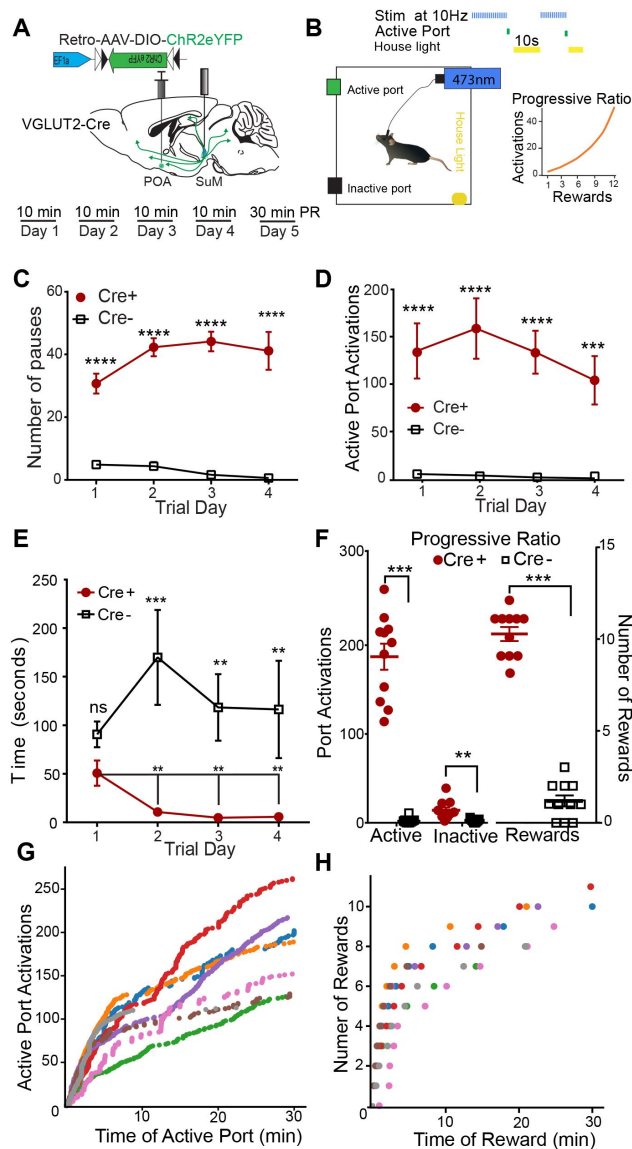


Figure 6.

SuM^{VGLUT2+}::POA neurons drive can drive instrumental actionoutcome operant behavior

(A) Schematic of injection and implant in VGLUT2+ Cre mice and paradigm of testing in 10-minute trials with 4 days before a progressive ratio (PR) trial on day 5. (B) Illustration of the testing paradigm and set up. 10 Hz photostimulation was applied during the trials. Activation of the active port trigger the house light and paused stimulation for 10 seconds. Also shown is the progressive ratio used with number of required port activations per reward on vertical axis and reward number on the horizontal. (C) Cre+ mice (n=11) activated the active port triggering significantly (****p<0.0001) more pauses in stimulation than Cre- mice (n=11) mice on all 4 days of testing. (D) Cre+ mice activated the port significantly (****p<0.0001, ***p<0.001) than the Cre- mice on all 4 trials. Cre+ activated the active port significantly (p<0.001) on during the fourth trial than the first. (E) The time to first activation of the active port was not significantly (p=0.32) different during the first trial but Cre+ mice became significantly (**p<0.01, ***p<0.001) on subsequent trials. (F) On the fifth day after four 10-minute trials mice were tested for 30 minutes using a progressive ratio. Cre+ performed significantly (p<0.0001) more active port activations that Cre- mice and activated the inactive port significantly more (p<0.005) times. Cre+ mice also triggered significantly (p<0.001) more pauses in photostimulation that Cre- mice. (G) Individual data for a representative cohort of Cre+ mice showing cumulative active port activations by Cre+ mice (n=7) as function of time during the progressive ratio test are shown illustrated ongoing engagement of the active port throughout the 30-minute trial. (H) The cumulative pauses in photostimulation (rewards) earned as a function of time during the progressive ratio trial are show for individual Cre+ mice. Mice earned between 7 and 11 pause reward during the trial. All data plotted as means $\pm$ SEM.

Analyses of port activations during the 10-min trials revealed that Cre⁺ mice (n = 11) paused the photostimulation (reward) significantly ($p < 0.0001$) more on all four days of testing compared to Cre⁻ (n = 11) mice (**Figure 6C**). The number of 10-second pauses approached the maximum, 60 possible in a 10-min period. Cre⁺ mice engaged the active port significantly ($p < 0.001$) more than Cre⁻ mice (**Figure 6D**). Indicating no difference in baseline performance of the task, the time to the first engagement of the active port was not significantly ($p = 0.32$) different for Cre⁺ compared to Cre⁻ mice during the first test trial. During the three subsequent tests Cre⁺ mice were significantly ($p < 0.01$) faster to engage the active port after the start of the trial compared to Cre⁻ mice. Cre⁺ mice also were significantly ($p < 0.01$ paired t-test) faster in trials two through four at engaging the active port compared the first trial. Interestingly, Cre⁺ mice engaged the active port significantly ($p = 0.009$ paired t-test) fewer times on the fourth day than the first day but achieved more pauses (**Figure 6E**).

To examine effort-related motivation we employed a 30-min test using a progressive ratio requiring exponentially increasing number of active port activations to trigger a pause in the photostimulation on the fifth day of operant behavior testing (**Figure 6B**) (63). During trials using a progressive ratio, Cre⁺ mice activated the active port and paused stimulation significantly ($p < 0.001$) more times than Cre⁻ mice (**Figure 6F**). Examination of individual cumulative active port activation data showed Cre⁺ mice continue to engage the active port throughout progressive ratio trials despite the increasing work required to generate each pause in photostimulation. No breakpoint was observed during the trial (**Figure 6G and H**). This finding indicates that activation of SuM^{VGLUT2+}::POA neurons remains salient and motivating through the tested period and that photostimulation of does not evoke behaviors precluding completion of the task. Examination of reward behavioral epochs for representative animals shows that for lowest total number of rewards earned (green dots) the next pause required 20 activations of the active port and for the highest (red dots) the next pause required 50 activations of the active port. Taken together, results from instrumental reinforcement tasks demonstrate that activation of SuM^{VGLUT2+}::POA neurons does not solely evoke innate stereotyped behaviors and can drive active coping in the form of instrumental behaviors.

SuM^{VGLUT2+}::POA neurons are recruited during active coping behaviors in forced swim testing

The forced swim test is increasingly understood as an assay of coping strategy with passive and active components (64, 65). During a first trial of forced swimming mice shifted from active escape behaviors (wall climbing, robust swimming) to in passive (immobile floating) behaviors. We found that SuM^{VGLUT2+}::POA neurons were recruited in a short dunk assay (**Figure 3F and G**), which elicited active escape behaviors (**Supplemental Figure 4A**). To test if recruitment of SuM^{VGLUT2+}::POA neurons changes with the shift from active to passive behaviors during a longer swim assay we carried out experiments using fiber photometry paired with automated computer-based behavior analysis (66). The automated behavior scoring enabled high temporal resolution analysis of behaviors (**Supplemental Video 4**) and each frame of the video was coded for the classified behavior and plotted as color coded point. We observed 4 types of behavior during forced swimming: climbing, swimming, hindpaw swimming, and immobile floating and trained a model to identify each. Analysis of behavior showed that after the first minutes of the trial mice shifted from swimming and climbing to immobile floating with intermittent swinging or hindpaw swimming. We examined if recruitment of SuM^{VGLUT2+}::POA neurons changed in correlation with this shift to more passive behavior. As shown in representative trace (**Figure 7A**) the Ca²⁺-dependent signal decreased the shift in behavior was occurring. To combine behavioral data we assigned numerical values, one through four, to each behavior: climbing (4), swimming (3), hindpaw swimming (2), and immobile floating (1). The mean was plotted as a heat map (red higher values to blue lower values) to be compared to time locked fiber photometry data (**Figure 7B**). Examining the timeframe around the behavioral shift in each animal we found that as behavior shifted from active coping (climbing, swimming) to more passive strategies and

greater immobility the Ca^{2+} signal substantially declined (**Figure 7B**). Examining data after this initial transition we observed that many elevations in the Z-score were accompanied by changes in behavior to swimming or hindpaw swimming (**Figure 7A**). To further examine this possible correlation, we examined the time frame around transition to hindpaw swimming for changes in Ca^{2+} signal. In a 20-sec window centered on the onset of hindpaw swimming we hind swimming was accompanied by a rise in the Ca^{2+} dependent signal across multiple events within a trial (**Figure 7C**). A similar analysis across events in multiple animals demonstrated a similar rise in the Ca^{2+} dependent signal from $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons during the shift from immobility to hindpaw swimming or swimming behaviors (**Figure 7D**). Time series analysis of 20-sec epochs centered on random time intervals did not reveal any rise in the Ca^{2+} dependent signal (**Figure 7E**). These data indicate that recruitment of $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons fluctuates with changes in coping strategy during a forced swim assay with decreased engagement of this circuit during times of immobility.

SuM^{VGLUT2+}::POA neurons promote active coping during forced swim test

We found that $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons are activated by acute stressors (**Figure 3**), are recruited during times of greater active coping behaviors (**Figure 7**). To test if $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons can drive a switch in coping strategy in the context of ongoing stressor we used a two-day forced swim stress test (65, 67). We tested animals during the second day, when immobile floating is the predominant behavior. As in previous experiments, we used VGLUT2-Cre (Cre+) or WT (Cre-) mice injected in the POA bilaterally with Retro-AVV-DIO-ChR2eYFP. On day one we subjected mice to 15 mins of forced swim. On the subsequent day we repeated the forced swim for six mins divided into three periods, pre-stimulation, stimulation at 10Hz, and post-stimulation (**Figure 7F**). Trials on the second day were recorded and scored by a blinded observer for time spent immobile in each 2-min period. Example of Cre+ mouse with stimulation 10Hz is shown in **Supplemental Video 3**. During the pre-stimulation period Cre+ (n=9) and Cre- (n=11) mice there was not a significantly ($p=0.45$) different amount of time immobile. During the stimulation period Cre+ mice began swimming and attempting to climb the wall of the circular swim arena, reflected by significantly ($p<0.0001$) less time immobile (**Figure 7H**). Interestingly, in the post stimulation phase the difference in time spent immobile between Cre+ can WT mice decreased but remained significantly ($p=0.016$) different. These results indicate that in the context of an ongoing stressor activation of $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons is sufficient to trigger a change in coping strategy from a largely passive (floating) to and active (swimming, climbing). The persistent effect of the acute activation into the poststimulation phase suggests that activation of $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons may shift how the stressor is processed or approached.

We next examined the effect of 10 Hz photostimulation of SuM^{VGAT} neurons on coping strategy using the same two-day swim paradigm. VGAT-Cre mice were injected in the SuM to express ChR2eYFP (n=10), or control (eYFP) (n=11) and a fiber optic was placed over SuM (**Figure 7F**). Quantification of behavior revealed no significant difference in time spent immobile during the pre-stimulation or the stimulation periods. Surprisingly, in the post-stimulation period we observed a dramatic shift in behavior marked by significant ($p=0.0004$) decrease in time spent immobile (**Figure 7I**). The amplitude of change in behavior was similar as we observed during the stimulation phase of the experiments on $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons (**Figure 7H**). One interpretation of these data is that release of sustained local inhibition leads to rebound activity of output $\text{SuM}^{\text{VGLUT2}^{+}}::\text{POA}$ neurons.

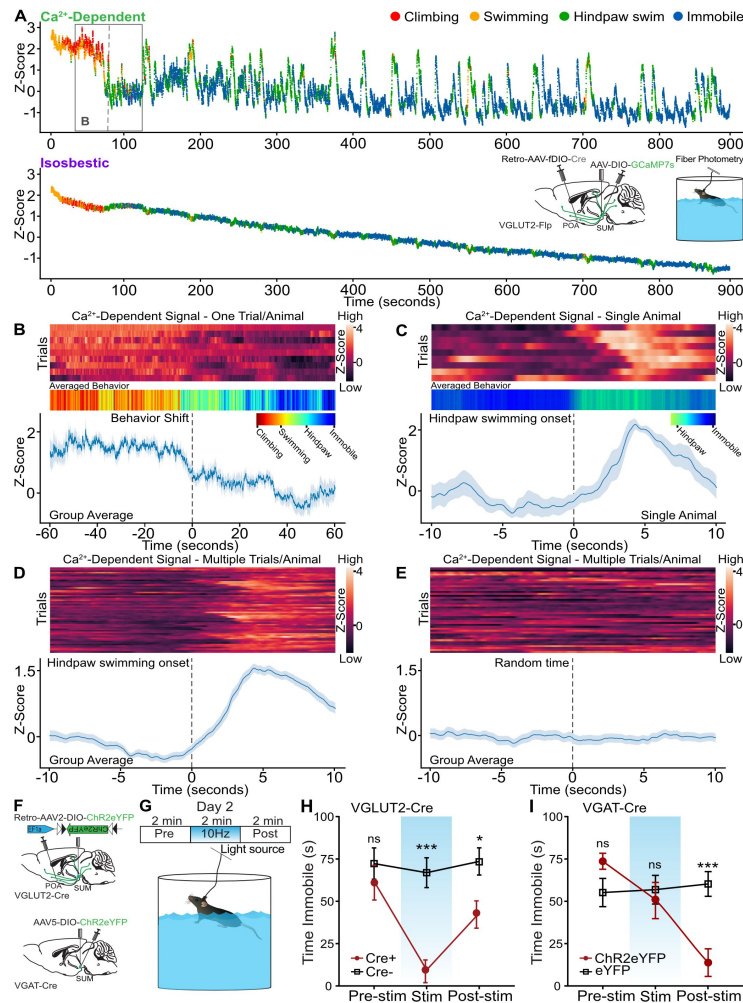


Figure 7.

SuM^{VGLUT2+}::POA neurons are recruited during active coping and activation is sufficient drive a switch to active coping.

(A) Color coded per video frame (dot color) behavior combined with normalized Ca^{2+} dependent and isosbestic fiber photometry signals. Scoring and analysis of climbing, swimming, hindpaw swimming, and immobility was completed using deep learning based classification and quantification (LabGym) of recorded behavior. Inset shows injection and implants and behavioral assay. Box highlights the time frame shown in B. (B) Heat maps for recordings obtained mice ($n=9$) during forced swim, averaged color-coded behaviors, and mean $\pm$ SEM Z-score of Ca^{2+} dependent signal for 2-min period around behavioral shift from climbing and swimming to hindpaw swimming and immobility. (C) Representative heat map, averaged behavior, and mean $\pm$ SEM Z-score of Ca^{2+} dependent signal for 20-second window around onset of hindpaw swim from a representative animal. (D) Representative heat map and mean $\pm$ SEM Z-score of Ca^{2+} dependent signal for 20-second window around onset of hindpaw swim from for events from nine mice. Engaging in hindpaw swim correlates with brief increase in Ca^{2+} signal. (E) Representative heat map and mean $\pm$ SEM Z-score of Ca^{2+} dependent signal for 20-second window around random time points. (F) Schematic of injection and implant in VGLUT2-Cre or VGAT-Cre mice. (G) Illustration of testing paradigm on second of forced swim test following 15-minute swim on first day. (H) The average time spent immobile during the pre-stimulation period was not significantly ($p=0.48$) different. During 10 Hz photostimulation VGLUT2-Cre+ mice engaged in vigorous swimming and the time spent immobile was significantly ($***p<0.001$) less than Cre- mice. In the post-stimulation period, the time spent immobile remained significantly ($*p=0.016$) lower in Cre+ mice compared to Cre-. (I) In VGAT-Cre+ mice expressing ChR2 ($n=10$), the average time spent immobile during the pre-stimulation period was not significantly ($p=0.08$) different compared to eYFP controls. Photostimulation did not significantly increase time immobile ($p=0.64$). However, there was a significant ($***p=0.0004$) decrease in the time spent immobile compared in the post-stimulation period.

Suppression of SuM^{VGLUT2+}::POA neurons is required for feeding

Feeding and responding to threats are conflicting actions and SuM^{VGLUT2+}::POA neurons may play a role in switching between behavioral paradigms (e.g., feeding vs escape). We have found threatening stressors recruit SuM^{VGLUT2+}::POA neurons. Responding to an acute threatening stressor would prevent feeding. We sought to examine SuM^{VGLUT2+}::POA neuron activity in relation to feeding using fiber photometry. To promote varying drive for food mice were given *ad lib* access to food (fed condition) or food deprivation for 24 hours and on testing day presented with a chow pellet while Ca²⁺ dependent and isosbestic signals were recorded. In the fed state mice spent significantly ($p < 0.001$) less time interacting with the food (**Figure 8C**) and ate significantly ($p < 0.001$) less (**Figure 8D**) compared to food deprived state. We analyzed the Ca²⁺ dependent and isosbestic signals around presentation of food to fed and food deprived mice. In the food deprived state mice spent more time with the food and ate more food (**Figure 8E and F**). At the time of food presentation, the Ca²⁺ dependent signal decreased but not the isosbestic signal. The change in the Ca²⁺ dependent was larger in the food deprived state (**Figure 8E and F**). Together this data supports suppression of SuM^{VGLUT2+}::POA neural activity during consummatory behavior with greater consumption associated with greater suppression.

To examine whether activation of SuM^{VGLUT2+}::POA neurons disrupted consummatory behavior we tested the impact of photostimulation on feeding behavior in food deprived mice. Using VGLUT2-Cre (Cre+) or WT (Cre-) mice injected with Retro-AAV5-DIO-ChR2eYFP into the POA and implanted with a fiber optic midline over SuM we examined how feeding behaviors in food deprived mice were altered by 10 Hz photostimulation of SuM^{VGLUT2+}::POA neurons (**Figure 8G and D**). We used trials lasting 20 mins with unrestricted access to food added at the start of the trial. Each trial was divided into three periods, a 5-min pre-simulation period, 10-min stimulation period, and a 5-min post-stimulation period. Trials were recorded and scored for time spent interacting with the food pellet. During the pre-simulation period Cre+ ($n=17$) and Cre- ($n=16$) mice spent interacting with the food was not significantly ($p=0.37$) different. With the start of photostimulation the Cre+ mice stopped eating and interacting with the food. During the stimulation period Cre+ mice spent significantly ($p=0.0002$) less time interacting with the food compared to Cre- mice. Surprisingly, the decrease in time spent interacting with food continued into the post-stimulation phase when Cre+ mice interacted with the food significantly ($p < 0.0001$) less (**Figure 8E**). Reflecting the decrease in time spent with the food, Cre+ mice consumed significantly ($p=0.001$) less food during the 20-min trial than Cre-mice. These results indicate that SuM^{VGLUT2+}::POA neurons can redirect behavior away from consumption and suppression of SuM^{VGLUT2+}::POA neurons is required for feeding even in a food deprived state. Ethologically, animals must choose rapidly between responding to threats and feeding when foraging. The findings here implicate this pathway in control of switching between these behaviors.

Discussion

The presented data show SuM^{VGLUT2+}::POA population of neurons have broad arborizations to brain regions involved in responding to threat and stress (**Figures 1**, **2**, and **Supplemental Figure 2**), are recruited by threatening stressors (**Figures 3** and **7**), can contribute to multiple aspects of behavioral responses to threatening stressors, and promote active coping behaviors (**Figures 6** and **7**). We also found that SuM^{VGLUT2+}::POA can drive aversion and although aversive stimuli can promote anxiety-like behaviors, photostimulation of SuM^{VGLUT2+}::POA stimulation did not (**Figure 5**). This finding suggests SuM^{VGLUT2+}::POA neurons may play a role in suppressing anxiety and passive fear behaviors to promote active engagement. Finally, examining the conflicting behaviors required for threat response and feeding we SuM^{VGLUT2+}::POA were suppressed during feeding and that activation of SuM^{VGLUT2+}::POA neurons blocked feeding activity both during stimulation in the period after stimulation (**Figure 8**). SuM^{VGLUT2+}::POA neurons are distinct subset neurons projecting to many but not all areas

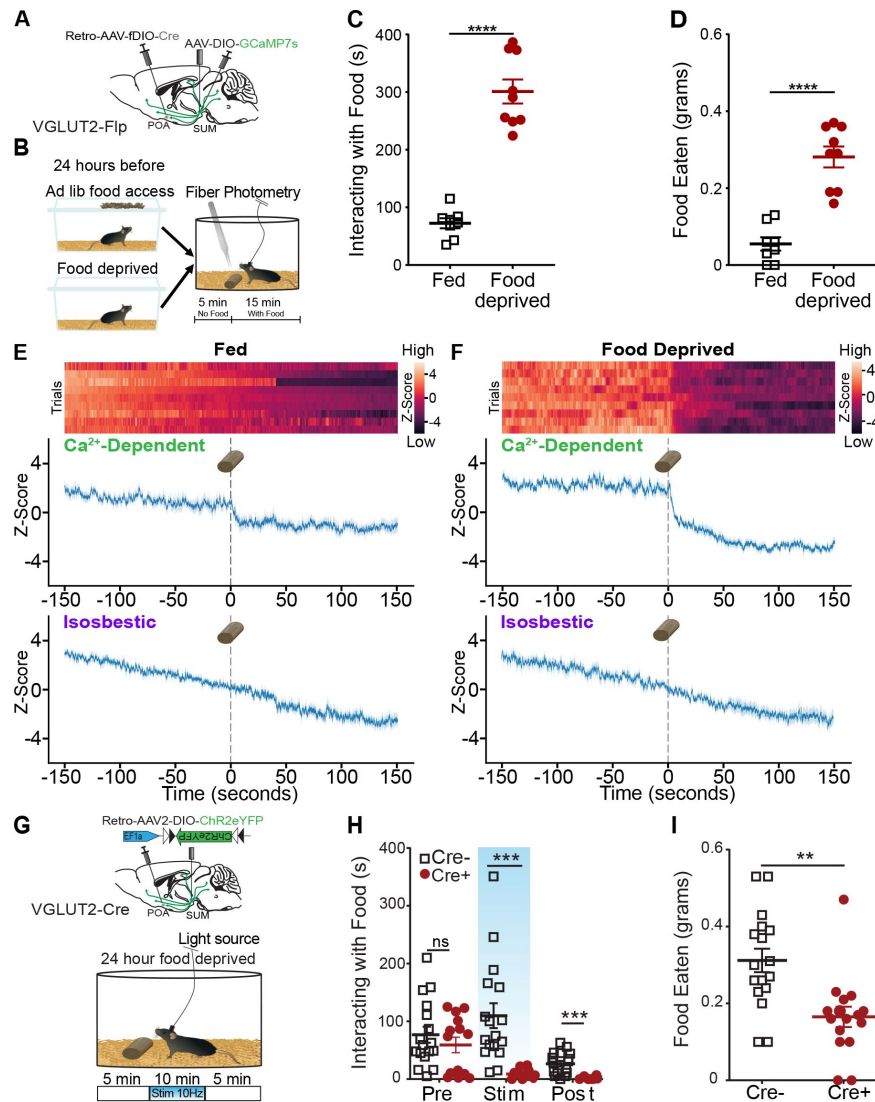


Figure 8.

Consummatory behavior suppresses SuM^{VGLUT2+::POA} neuron activity and SuM^{VGLUT2+::POA} neuron blocks food consumption.

(A) Schematic of injections in VGLUT2-Flp mice with AAV-DIO-GCaMP7s in SuM and Retro-AAV-fDIO-Cre in the POA and a fiber placed over SuM. (B) Mice were given ad lib food access or food deprived for 24 hours prior to testing. After being placed in arena and allowed to habituate mice were given a 20-minute trial divided into 5 mins baseline and 15 mins after a chow pellet was added to the arena. (C) Food deprived mice (n=9) spent significantly (p<0.001) more time interacting with the food pellet and (D) ate significantly (p<0.001) more than in the fed state. (E) Heat map and mean ±SEM z-scores for Ca²⁺ dependent and isosbestic signals for recordings obtained from 9 animals show a small drop in (E) fed mice state compared to a sharp decrease in Ca²⁺ activity after the introduction of the food pellet following addition in (F) food deprivation. (G) Mice were fasted for 24 hours prior to testing and the schematic 20 min paradigm was used. Following food deprivation animals were given access to a chow pellet. The trial was divided into a 5-minute pre- and post-stimulation periods with 10-minute period of stimulation. (H) The time spent interacting with the food was quantified for each period. Cre+ (n=16) and Cre- (n=17) mice both rapidly engaged with the food pellet and the average time spent interacting was not significantly (p=0.37) different. During the stimulation period and the post stimulation period the average time interacting with the food was significantly (p=0.0002, p=0.0001) lower in Cre+ mice compared to Cre- mice. (I) The food eaten during the total trial was calculated based on pellet weights and on average Cre+ mice ate significantly (p=0.001) less food during the trial than Cre- mice. (**p=0.01, ***p<0.001, ****p<0.0001)

receiving projections from the total SuM^{VGLUT2+} population. Notably, mHb and DG projections were present in the SuM^{VGLUT2+} population but absent in the SuM^{VGLUT2+}::POA projections (**Figure 3K**), consistent with a separation of DG and CA2 projecting SuM neurons (68). The extensive arborization of SuM^{VGLUT2+}::POA neurons challenges the often-employed neural circuit concept that projection from a single area to a second to mediate or modify a behavior. The projections to PVT, Hb, and PAG are of particular interest because these areas regulate stress coping (69–72). Dopamine receptor 2 expressing neurons in the PVT have been implicated in promoting active coping behaviors and may be a potential target of SuM^{VGLUT2+}::POA neurons (8, 9). The habenula also regulates active vs passive coping behaviors, and the lHb functions in aversive processing making the selective projections of SuM^{VGLUT2+}::POA neurons to the lHb of similar interest (69, 73–76). In addition, the PAG, which receives projections from SuM^{VGLUT2+}::POA neurons, has been implicated in regulating defensive behaviors and coping strategies (70, 71, 77). The arborization to these and other areas position SuM^{VGLUT2+}::POA neurons to be a central hub for regulating behavioral responses to threatening stressors by concurrent recruitment of multiple areas.

Superficially, the aversiveness of SuM^{VGLUT2+}::POA neuron stimulation we report contrasts with a recent study that found photostimulation of SuM neurons projections could be reinforcing. Photostimulation of projections from SuM to the septum drove place preference (septum) but, the same report found photostimulation of SuM projections in the PVT was aversive (19). We report here that SuM^{VGLUT2+}::POA neurons also project to the PVT and the septum (**Figure 2**). A plausible interpretation is that SuM contains separable molecularly defined populations of neurons which gate a host of distinct behavioral strategies through their connectivity and or co-transmitter content. Delineating the overarching circuitry will require further studies, but our results together with recent studies establish SuM is an important and poorly understood node for regulating appetitive and aversive motivated behaviors.

We addressed the challenge of potential confounds from locomotor activity inherent in active coping behaviors, which is of special concern here because a population of SuM neurons have been found to correlate with velocity of movement, by using multiple behavioral assays with fiber photometry and photostimulation. We found multiple threatening stressors (dunk, shock, ambush) with differing evoked behaviors all recruit SuM^{VGLUT2+}::POA neurons (**Figure 3**). We also found increased recruitment of SuM^{VGLUT2+}::POA neurons during active coping behaviors (**Figures 3 and 7**). We did not find evidence for recruitment of SuM^{VGLUT2+}::POA neurons during elevated locomotion in two cohorts in assays of free movement (**Figure 3K–M** and **Supplemental Figure 5**). These data are in contrast to recent studies that found activity of some neurons in SuM correlated future velocity (41). We carefully examined the behaviors elicited by activation of SuM^{VGLUT2+}::POA neurons both in absence of stressors (**Figure 4**) and during ongoing stressors (**Figure 7**) and found stimulation of SuM^{VGLUT2+}::POA neurons promoted active coping behaviors. Because escape behaviors could be innate fixed behaviors we examined if SuM^{VGLUT2+}::POA neurons could also drive performance of an operant task, which would be impaired if activation of SuM^{VGLUT2+}::POA neurons lead to fixed escape behaviors. We found mice effectively performed the operant task. In trials using a progressive ratio mice performed the task repetitively during sustained photostimulation because the number of port activations required to terminate the stimulation was high. Taken together the results from the behavioral tests indicate that SuM^{VGLUT2+}::POA promote flexible stressor appropriate coping behaviors and not generalized locomotion.

Although neurons that release both GABA and glutamate projecting to dentate gyrus are present in SuM, our anatomic studies indicate and SuM^{VGLUT2+}::POA neurons were discrete from GABAergic neurons do not project the dentate gyrus (**Figure 2** and **Supplemental Figure 2**) (21, 40). SuM^{VGAT} could be a functionally distinct population. We tested the effects of photostimulation of SuM^{VGAT} neuron and in contrast to SuM^{VGLUT2+}::POA neurons, found no effect on place aversion, minor decreases in locomotion, and no significant change in active coping behaviors during stress

(FST) during stimulation. Surprisingly, immediately after photostimulation stopped active coping behavior increased dramatically (**Figure 7I**). These data demonstrate functional separation of SuM^{VGAT} and suggest connection between the $\text{SuM}^{\text{VGLUT2+::POA}}$ and $\text{SuM}^{\text{VGAT+}}$ populations.

The aversiveness of $\text{SuM}^{\text{VGLUT2+::POA}}$ neuron stimulation we report contrasts with a recent study that found photostimulation of SuM neurons projections could be reinforcing. However, more closer examination suggests the potential underlying diversity of SuM neurons may explain the results. Photostimulation of projections from SuM to the septum drove place preference but photostimulation of SuM projections in the PVT was aversive (19). We report here that $\text{SuM}^{\text{VGLUT2+::POA}}$ neurons also project to the PVT and the septum (**Figure 2**). A plausible interpretation is that SuM contains separable populations. One set (PVT projecting) could drive aversion and the other (non-PVT projecting) may regulate reinforcement. Delineating the overarching circuitry will require further studies. Our results together with recent studies establish SuM is an important and poorly understood node for regulating appetitive and aversive motivated behaviors.

In conclusion, $\text{SuM}^{\text{VGLUT2+::POA}}$ neurons arborize to multiple areas involved in stress and threat response that promote active coping behaviors. Passive coping strategies have been associated with unescapable stress, anhedonia, and depression (7, 78–80). Targeting neuromodulation of a circuit able to act across many brain areas may represent a therapeutic avenue for common psychiatric conditions and $\text{SuM}^{\text{VGLUT2+::POA}}$ neurons are a newly identified node in critical approach-avoidance circuitry.

Acknowledgements

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Materials and Methods

Methods

Key Resources Table

Reagent type (species) or resource	Source or reference	Identifiers
Antibodies		
Alexa Fluor 488 Goat anti-Rabbit IgG	Invitrogen	Cat# A11008
Phospho-c-Fos (Ser32) (D82C12) XP®	Cell Signaling Technology	Cat# 5348
Chemicals		
4% paraformaldehyde	J.T. Baker, Avantor	S898-09
Goat Serum	Sigma Aldrich	G9023
Vectashield Hardset Antifade Mounting Medium With DAPI	Vector Laboratories	Cat# NC9029229
Experimental Models: Organisms/Strains		
Slc17a6 ^{tm2(cre)Lowl/J} (VGLUT2-Cre)	The Jackson Laboratory	RRID:IMSR_JAX:016963
B6;129S-Slc17a6 ^{tm1.1(flpo)Hze/J} (VGLUT2-Flp)	The Jackson Laboratory	RRID:IMSR_JAX:030212
Mouse:B6.Cg-Gt(ROSA)26Sor ^{tm14(CAG-tdTomato)Hze/J} (Ai14)	The Jackson Laboratory	RRID:IMSR_JAX:007914
B6J.129S6(FVB)-Slc32a1 ^{tm2(cre)Lowl/MwarJ} (VGAT-Cre)	The Jackson Laboratory	RRID:IMSR_JAX:028862
C57BL/6J	The Jackson Laboratory	RRID:IMSR_JAX:000664
Bacterial and Virus Strains		
AAV5-EF1a-DIO-hChR2(H134R)-EYFP (2.5 × 10 ¹³ vg/ml)	Washington University Hope Center Viral Vector Core	N/A
AAV2-Retro-DIO-ChR2-eYFP (2.8 × 10 ¹² vg/ml)	Washington University Hope Center Viral Vector Core	N/A
AAV2-retro-FLEX-tdTomato (7 × 10 ¹² vg/ml)	Addgene	RRID: Addgene_28306-AAVrg
AAV5-DIO-ChR2-eYFP (1.4 × 10 ¹³ vg/ml)	Washington University Hope Center Viral Vector Core	N/A
AAV2-Retro-EF1a-fDIO-cre (7 × 10 ¹² vg/ml)	Addgene	RRID: Addgene_121675-AAVrg

AAV2-EF1a-Nuc-flox(mCherry)-EGFP (5 × 10 ¹² vg/ml)	Addgene	RRID: Addgene_112677-AAV5
Retro-AAV2-EF1a-Nuc-flox(mCherry)-EGFP	Addgene	RRID: Addgene_112677-AAVrg
AAV2-retro-EF1a-DIO-eYFP (3 × 10 ¹³ vg/ml)	Washington University Hope Center Viral Vector Core	N/A
AAV-retro-EF1a-Flpo (1.02 × 10 ¹³ GC/mL or 7 × 10 ¹² vg/ml)	Addgene	RRID: Addgene_55637-AAVrg
AAV9-EF1a-fDIO-cre (2.5 × 10 ¹³ GC/mL or 1 × 10 ¹³ vg/ml)	Addgene	RRID: Addgene_121675-AAV9
Retro-AAV2-EF1a-Cre (2.1 × 10 ¹³ GC/mL)	Addgene	RRID: Addgene_55636-AAVrg
AAV9-syn-FLEX-GCaMP7s-WPRE	Addgene	RRID: Addgene_104487-AAV9
Software		
Bonsai	Bonsai-rx.org	N/A
Ethovision	Noldus	N/A
GraphPad Prism	GraphPad Software	N/A

For further information regarding reagents and resources, contact Aaron Norris, norrisa@wustl.edu.

Experimental model and subject details

Adult (25-35 g, at least 8 weeks of age upon experimental use) male and female VGLUT-Cre (RRID: IMSR_JAX:016963), VGLUT2-Flp (RRID: IMSR_JAX: 030212), Ai14 (RRID: IMSR_JAX: 007908), VGAT-Cre (RRID: IMSR_JAX: 028862) and C57BL/6J (RRID: IMSR_JAX: 000664) mice (species *Mus musculus*) were group housed (no more than five littermates per cage) in a 12 hr:12 hr light:dark cycle room with food and water ad libitum⁽⁸¹⁾. The Washington University Animal Care and Use Committee approved all procedures which adhered to NIH guidelines.

Stereotaxic surgery

Injections and implantations were done as described previously^(34, 82). Briefly, in an induction chamber, mice were anesthetized (4% isoflurane) before being placed in a stereotaxic frame (Kopf Instruments). Anesthesia was maintained with 2% isoflurane. Mice were then injected unilaterally or bilaterally, depending on the combination of virus(es) used and brain regions injected. A blunt needle Neuros Syringe (65457-01, Hamilton Con.) and syringe pump (World Precision Instruments) were used to perform the injection schemes below. After surgery, a warmed recovery chamber housed the animal while it recovered from anesthesia before being returned to its home cage.

Virus	Virus Volume	Brain region/coordinates
AAV5-EF1a-DIO-hChR2(H134R)-EYFP	100 nl	Region, laterality, (AP , ML , DV)
AAV2-retro-DIO-ChR2-eYFP	100 nl	SuM (AP -2.7, ML +0.05, DV -4.3)
AAV2-retro-FLEX-tdTomato	100 nl	Septum (AP +0.9, ML +0.3, DV -3.5) LPAG, unilateral (AP -4.65, ML +0.6, DV -2.85)
AAV2-retro-EF1a-fDIO-cre	100 nl	POA, bilateral (AP +0.45, ML $\pm$ 0.5, DV -5.35)
AAV2-EF1a-Nuc-flox(mCherry)-EGFP	300 nl	SuM (AP -2.7, ML +0.05, DV -4.3)
AAV2-retro-EF1a-Nuc-flox(mCherry)-EGFP	150 nl	POA, bilateral (AP +0.45, ML $\pm$ 0.5, DV -5.35)
AAV22-retro-EF1a-DIO-eYFP	100 nl	Acb (AP +1.4, +0.6 ML, DV -4.75) POA, bilateral (AP +0.45, ML $\pm$ 0.5, DV -5.35)
AAV2-retro-EF1a-Flpo	50 nl	POA, bilateral (AP +0.45, ML $\pm$ 0.5, DV -5.35)
AAV9-EF1a-fDIO-cre	100 nl or 50 nl	SuM m(AP -2.7, ML +0.05, DV -4.3)

Injections were made at a rate of 50 nl/min, with the injection needle being withdrawn 5 min after the end of the infusion. Fiber optics for photostimulation or optical fibers for fiber photometry were implanted after injections for all behavioral experiments. Fiber optics for photostimulation or optical fibers for fiber photometry were implanted after injections for all behavioral experiments.

Fiber optic implants for photostimulation were fabricated as previously described using 200 μ m glass fibers and implanted midline over SuM (34, 83). For implantation, the skull was cleaned and etched with OptiBond® (Kerr) and the fiber was cemented to the skull with Tetric N-Flow® (Ivoclar Vivadent). Blue light was used to cure and harden cement. Mice were allowed to recover for at least seven days before the start of behavioral experiments. The same process was used for implantation for fiber photometry fibers, which were trimmed to length (NeuroPhotometrics).

Anatomical tracing

For anterograde viral tracing experiments, virus (was injected at least six weeks prior to transcardial perfusions with 4% paraformaldehyde to allow for anterograde transport of the fluorophore. AAV5-EF1a-DIO-hChR2eYFP or AAV5-EF1a-DIO-eYFP were used. Alternatively, to label only SuM^{VGLUT2+} POA neurons for anterograde tracing Retro-AAV2-DIO-eYFP or Retro-DIO-ChR21eYFP was injected into the POA with AAV-fDIO-Cre injected in the SuM. A minimum of 6 weeks was allowed prior to sacrifice, harvesting or brains, and sectioning (30 μ M). Serial 30 μ M sections approximately 60 μ M apart were examined. For retrograde studies viruses were injected (see figure legends and text for specific viruses) at the targeted site and table for specific viruses(84-86). Three weeks were allowed to elapse prior to harvesting brains following injections. Images were obtained on a Leica DM6 B upright microscope and processed using Thunder imaging station (Leica).

Brain clearing and light sheet microscopy

Tissue clearing and imaging was carried out on brains collected and fixed in 4% PFA as described above by LifeCanvas Technologies. Briefly, brains were fixed using SHIELD post ix and cleared for 7 days in SmartClear II Pro. Index matched with EASYIndex at room temperature. Samples were mounted ventral side up and imaged at 3.6x with pixel size 1.8 x 1.8 mm, axial resolution < 4 μ m, z

step 4 μm in 488nm channel. Fos was labeled by Alexa Fluor 488. SuM boundaries were defined by -2.6 to -2.95 rostral to Bregma. The medial mammillary nucleus and the mammillary recess of the 3rd ventricle marked the medial and ventral boundaries while fornix marked the lateral, and fasciculus retroflexus the dorsal boundaries. Images were quantified by a trained laboratory member who was blind to the experimental conditions.

Immunohistochemistry

Mice were intracardially perfused with 4% PFA and then brains were sectioned (30 microns) and placed in 1x PB until immunostaining. Free-floating sections were washed in 3 times in 1x PBS for 10 min intervals. Sections were then placed in blocking buffer (0.5% Triton X-100% and 5% natural goat serum in 1x PBS) for 1 hr at room temperature. After blocking buffer, sections were placed in primary antibody rabbit Phospho-c-Fos (Ser32) antibody (1:500, Cell Signaling Technology) overnight 4 °C temperature. After 3 x 10 min 1x PBS washes, sections were incubated in secondary antibody goat anti-rabbit Alexa Fluor 488 (1:2500, Invitrogen) for 2 hr at room temperature. Sections were washed in 1x PBS (3 x 10 min) followed by 2 x 10 min 1x PB washes. After immunostaining, sections were mounted on Super Frost Plus slides (Fisher) and covered with Vectashield Hard set mounting medium with DAPI (RRID:AB_2336788, Vector Laboratories) and cover glass prior to being imaged.

Imaging and cell quantification

‘The Mouse Brain in Stereotaxic Coordinates’ provided the framework to label brain sections relative to bregma(87 [↗](#)). A Leica DM6 B epifluorescent microscope was used to image all sections. For eYFP visualization, a YFP filter cube (Excitation: 490-510, Dichroic: 515, Emission: 520-550) was used and for tdTomato visualization, a Texas Red Filter Cube (Excitation: BP 560/40, Dichroic: LP 585, Emission: BP 630/75) was used. Fos was labeled by Alexa Fluor 488. SuM boundaries were defined by -2.6 to -2.95 rostral to Bregma. The medial mammillary nucleus and the mammillary recess of the 3rd ventricle marked the medial and ventral boundaries while fornix marked the lateral, and fasciculus retroflexus the dorsal boundaries. Images were quantified by a trained laboratory member who was blind to the experimental conditions.

Forced-Swim Test

For stress induction via forced swim (**Figure 4** [↗](#))(65 [↗](#), 88 [↗](#)), mice were individually placed in a cylindrical container (18 cm in diameter) filled with water at 25 $\pm$ 1 °C for 15 min. Prior to stress force swimming, mice were habituated to the arenas for three days prior to the beginning of FST to minimize stress. Control mice were brought to the behavioral testing area but remained in the home cage until sacrifice. Water was replaced after every animal.

Fiber Photometry

For all fiber photometry experiments the same strategy to selectively label SuM^{VGLUT2+}::POA neurons was used. VGLUT2-Flp mice were injected bilaterally in POA with AAV-Retro-EF1a-fDIO-Cre and with AAV-syn-FLEX-GCaMP7s-WPRE in SuM. After two weeks, mice were implanted with fiber-optic cannulas (200 μm) in SuM (D/V -4.3-5)(89 [↗](#)). Mice recovered a minimum of one week prior to behavioral testing. Recording of Ca²⁺ dependent and isosbestic signals were obtained using previous described methods with Bonsai software and FP3002 (Neurophotometrics)(63 [↗](#), 90 [↗](#)). 470 and 415 nm LEDs were used to record interleaved isosbestic and Ca²⁺ dependent signals following the manufacturer directions.

For repeated forced swim experiments (dunk tank) mice were placed inside a custom-made plastic rectangular enclosure (20 cm x 20cm x 23 cm) filled with water 30 $\pm$ 1 °C. The enclosure has a square wire mesh platform that could be raised and lowered without touching the animal. The platform was lowered for the swim session and raised at the conclusion of each session. During

testing mice were tethered to a fiberoptic patch cabled that was attached to a counter balanced arm that prevented downward force on the animal. Time mobile was quantified by a trained observer for the ten 30s trials.

For foot shock stress testing, mice were individually placed inside a custom-made clear plastic box (15.24 L x 13.34 W x 14 H cm) inside a sound-attenuated cabinet. A speaker was placed 4 cm above the chamber for the delivery of auditory cues (75dB). A constant current aversive stimulator (ENV-414S) delivered foot shocks through a grid floor (0.7mA). Five shocks of 2 sec were delivered after a 30 s tone. Intertrial interval ranged from 90 to 180 sec. After the session, the animal was removed from the chamber and placed back in its home cage. The chamber and grid were wiped down with 70% ethanol between animals.

For open field testing for velocity measurements, mice were tethered to a fiberoptic patch cable and placed inside a custom-built square arena (50.8 cm x 50.8 cm x 50.8 cm). Mice were allowed to explore the arena for a 20 min session. Velocity was quantified using scripts in Bonsai 2.4. Bedding in the arena was replaced between animals, and the floors and walls of the arena were wiped down with 70% ethanol.

Similar to the repeated forced swim experiments, for forced swim mice were placed on top of a square wire mesh platform that could be raised and lowered inside a custom-made plastic rectangular enclosure (20 cm x 20 cm x 23 cm) filled with water at 25 +/- 1 °C. The platform was lowered at the beginning of the test for 15 min and raised at the conclusion of the test. During testing, mice were tethered to a fiberoptic patch cable that was attached to a counter balanced arm that prevented downward force on the animal. Water was replaced between animals and the enclosure was wiped down with 70% ethanol and rinsed with water. Time immobile, hindpaw swim, swimming, and climbing were quantified using LabGym (Hu et al., 2022).

To test the response to a natural stressor that mice can potentially encounter in their natural habitat (91 [DOI](#)). in SuM^{VGLUT2+::POA} neurons, a remote-controlled spider was used to simulate an ambush of a potential predator. The remote-controlled spider (17 cm x 16 cm; Amazon) was placed inside a polylactic acid (PLA) enclosure (19 cm x 20 cm x 23 cm) that was then placed inside a custom-built square arena (50.8 cm x 50.8 cm x 50.8 cm). Mice were tethered to a fiberoptic patch cable and placed inside the arena for 10 min. Baseline activity was recorded for 5 min before the ambush. After 5 min, the animal was ambushed once it moved in close proximity to the spider's enclosure (Supplemental video 4). Velocity was measured and analyzed using DeepLabCut (Mathis et al., 2019) within Bonsai. The arena and spider's enclosure were wiped down with 70% ethanol between animals.

For experiments that examined recruitment of SuM^{VGLUT2+::POA} neurons during consummatory behavior, mice were food deprived or given adlib access to food for 24 hours prior to testing. On test day, mice were placed inside an 18 cm diameter round arena and allowed to habituate for 30 min before they were tethered to the fiberoptic path cable. Once tethered, mice were placed inside the arena for 20 min. Baseline activity was recorded for 5 min before the introduction of the chow pellet and for 15 min after the chow pellet was introduced. Chow pellets were weighed before and after the 20-min trial. The difference is reported as food eaten. The same procedure was followed for control mice with the only difference being that these animals were not food deprived for 24 hours.

For fiber photometry data analysis, the interleaved isosbestic and Ca²⁺ dependent signals were recorded at 60 fps (30 fps each). Deinterleaved signals were analyzed using methods as previously reported (90 [DOI](#)). Briefly, raw signals were smoothed using a moving average, fitted with an exponential curve using a non-linear least squares function for baseline correction, signals were standardized using the mean value and standard deviation (Z-Score), the standardized isosbestic and Ca²⁺ signals were scaled a non-negative robust linear regression, and normalized dF/F was

calculated. In experiments shown in **figures 7A** and **8 E-F** there was sustained step drop evident in the Ca^{2+} dependent signal reflecting change in population activity because it was not seen in the isosbestic signal. The nature and duration of the change of the signal precluded fitting a curve to the Ca^{2+} dependent signal. In these cases, we show both the Ca^{2+} dependent and isosbestic signals. Z-scores were calculated without baseline correction for both Ca^{2+} dependent and isosbestic signals based on the variability in the baseline state.

Real-time place aversion testing

For real-time place preference testing with optogenetic photostimulation we used custom-made, unbiased, balanced two-compartment conditioning apparatus (52.5 x 25.5 x 25.5 cm) as described previously (34, 82, 92). Mice were tethered to a patch cable that allowed free access to the entire arena for 30 min. Entry into one compartment triggered photostimulation, 1Hz, 5Hz, 10Hz, or 20 Hz (473 nm laser, 10 ms pulse width) that persisted while the mouse remained in the light paired side. The side paired with photostimulation was counterbalanced across mice. Ordering was counterbalanced with respect to stimulation frequency and placement. Bedding in the behavior apparatus was replaced between every trial, and the floors and walls of the apparatus were wiped down with 70% ethanol. Time spent in each chamber and total distance traveled for the entire 30-min trial was measured using Ethovision 10 (Noldus Information Technologies).

Light /Dark choice

For light/dark choice the same areas as used for real time preference testing were modified and used as previously described (93). On the light side a small sport light was placed overhead, and the walls were covered with white laminated paper. Light levels on this side measure 580-590 lux. For the dark side, an infrared spotlight was placed over head, to allow for video tracking of the mice. The walls were covered with matte black plastic. Light levels in the center of the dark side measure 100-110 lux. Animals were recorded using a USB web cam without an infrared filter. For real-time aversion testing photostimulation was paired with the dark side of arena as described above. For anxiety-like behavior testing stimulation was provided uniformly during the trials. Time spent in each chamber and total distance traveled for the entire 30-min trial was measured using Ethovision 10.

Observational behavioral assay

To observe behaviors evoked by photostimulation of $\text{SuM}^{\text{VGLUT2+}}::\text{POA}$ neurons we mice were habituated for at least 3 days prior to testing to round (18 cm diameter) clear arena with counterbalanced optical commutators to minimize impact of the head tether on movement. Testing occurred after habitation and approximately 2 cm of bedding material was placed in the arena. Behavior was recorded from the side and scored by a blinded observer.

Negative reinforcer two nose port operant behavior testing

For operant behavior testing we used Med Associates mouse operant conditioning chamber with dual nose ports and house light as previously described (63). Briefly, mice were tethered via cantilevered counterweighted optical commutator to a laser light source. A five-day protocol call was used. Day one through four were 10-min trials. Photostimulation was provided at 10 Hz and activation of the active port resulted in a 10-second pause in the photostimulation and activation of the house light inside the arena. On the first four days each activation of the active port outside of a 10-second pause resulted in a new pause. On day five was a 30-min trial using a progressive ratio protocol the number activation of the active port to generate a pause increased with each activation based on the *number of activations* (j) = $[5e^{(0.2j)} - 5]$ round to the nearest integer generating the schedule 1, 2, 4, 6, 9, 12, 15, 20, 25, 32, 40, 50.... (94). The trial was limited to 30-mins due to concern for animal welfare due to the head tether and confined space. Photostimulation terminated at the conclusion of the trial.

Forced Swim with photostimulation

Forced swim trials with optogenetic stimulation were done as a two-day test. On the first day all mice were subjected to a 15-min forced swim, dried and returned to the home cage. On the second day they were tethered and subjected to a 6-min forced swim divided into three periods, each two mins in length, pre stimulation, stimulation, and post stimulation. 10 Hz photostimulation was provided during the trial. Mice were closely monitored during each trial of swimming. Trials were recorded and scored by a blinded observer for time spent immobile on the second day of the test.

Feeding after food deprivation

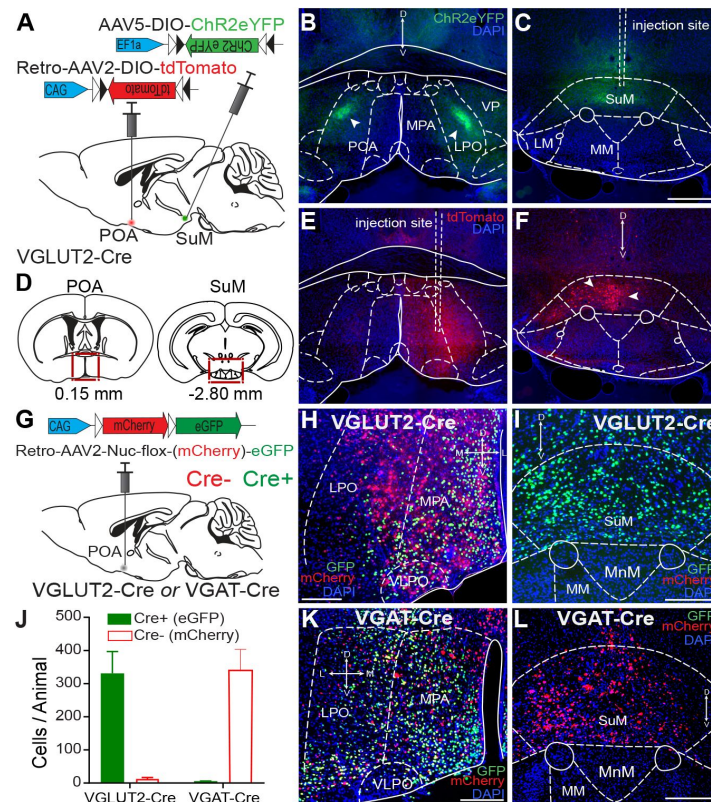
For tests involving brief access to food after deprivation, the same 18 cm diameter round arenas with counterbalanced optical commutators were used. Mice were habituated to the arenas for a minimum 3 days prior to testing. Mice were food deprived by removal of food from the home cage 24 hours prior to testing. Mice were placed in the arena and allowed to habituate prior to introduction of a chow pellet. The 20-minute trial with the food pellet was recorded and scored by a blinded observer. Chow pellets were weighed before and after the 20-min trial. The difference is reported as food eaten.

Open Field Test

For Open Field testing we used a purpose-built 20in square behavior arena. Mice were tethered to a patch cable and placed into the behavioral arena. The laser frequency was set to 10 Hz and was left on for 20 mins. Distance moved for the 20-min trial was quantified using Ethovision 10. Bedding in the arena was replaced between every trial, and the floors and walls of the arena were wiped down with 70% ethanol.

Statistical analyses

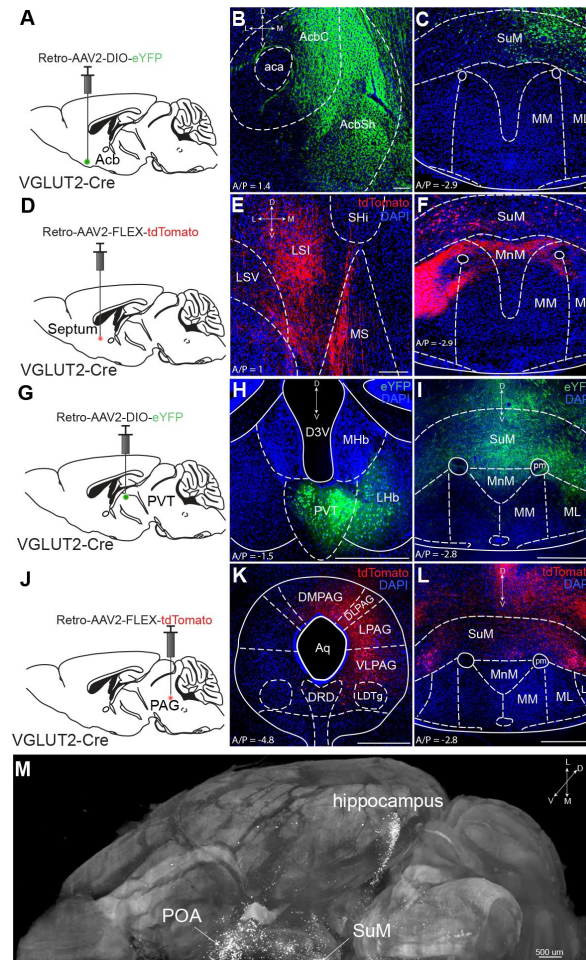
Statistical analyses were conducted using GraphPad Prism software. Data are shown as mean $\pm$ SEM except for z-scores which are shown as $\pm$ 95% confidence interval, as noted in text. Values for individual p values are given in the text and figure legends. Significance was held at α less than 0.05. Unless noted in the text, nonparametric Mann-Whitney tests were used for statistical comparisons. In cases of multiple comparisons, Holm-Šidák method was used to correct for multiple comparisons. Paired testing was used with comparing within cohorts with repeated measurements and unpaired between cohorts. All “n” values represent the number of animals in a particular group for an experiment. For fiber photometry statistical analysis, the mean signal of the baseline and stimulus windows was used, and comparisons were made using the Wilcoxon ranked-sum test, with $\alpha=0.05$.



Supplemental Figure 1.

SuMVGLUT2⁺ neurons project to the preoptic area of the hypothalamus and are not VGAT⁺.

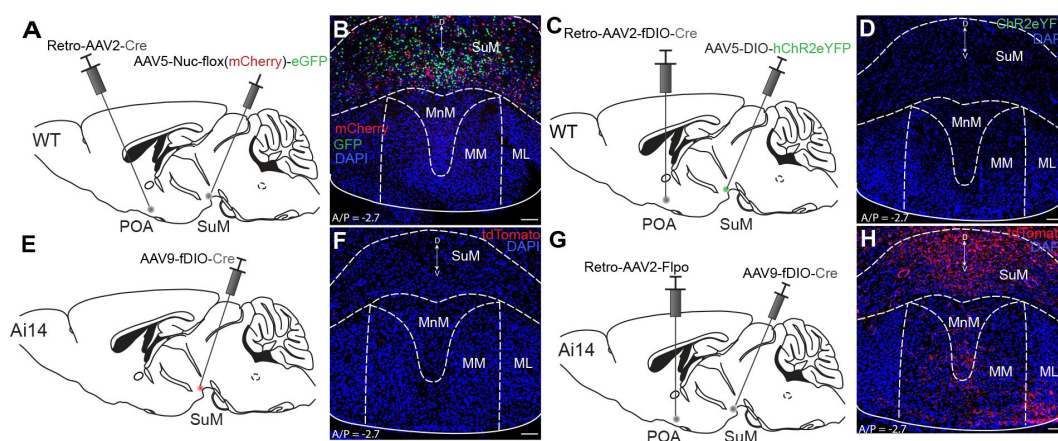
(A) Schematic of injections into VGLUT2-Cre mice of Retro-AAV-DIO-tdTomato into the POA and AAV-DIO-ChR2eYFP into the SuM. (B and C) Projections of VGLUT2+ SuM neurons and evident in the POA with dense labeling in LPO and the site of viral injection in SuM (500um scale bar). (D) The brain regions, POA and SuM, shown in (B,C, E and F). (E and F) Cell bodies of VGLUT2+ SuM neurons are evident in Sum with injection of Retro-AAVs in POA. (G) Schematic of injection in POA of fluorophore switching construct in Retro-AAV in VGLUT2-Cre or VGAT-Cre mice. (H and I) Labeling of Cre- (red) and Cre+ neurons in POA but only Cre+ seen in SuM of VGLUT2-Cre mice showing all SuM::POA neurons are VGLUT2+. (J) Quantification of labeling of Cre+ and Cre- cells in SuM following injections in POA in VGLUT2-Cre and VGAT-Cre mice. (K and L) Labeling of Cre- (red) and Cre+ neurons in POA but only Cre- seen in SuM of VGAT-Cre mice. (100um scale bar)



Supplemental Figure 2.

Retrograde verification of projection targets of SuM^{VGLUT2+}::POA.

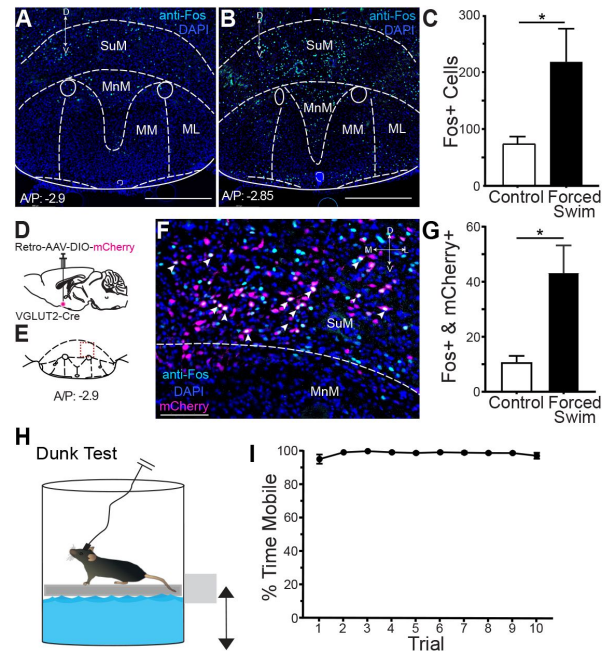
(A, D, G, and J) Schematic showing the injection sites of Retro-AAV's in VGLUT2-Cre mice. (B-C) In VGLUT2-Cre mice injection of Retro-AAV-DIO-eYFP unilaterally in the (B) accumbens nucleus resulted in labeling of cell bodies in (C) SuM. (E-F) Similarly, injection of Retro-AAV-DIO-tdTomato into (E) septum resulted in labeling of cell bodies in (F) SuM. (H-I) injection of Retro-AAV-DIO-eYFP targeting the (H) PV and LHb resulted in cell bodies in (I) SuM being labeled with eYFP. (K-L) Retro-AAV-DIO-tdTomato injected into (K) PAG labeled cells in (L) SuM. (M) The cleared brain hemisphere from a VGLUT2-Flp mouse injected in the POA with injected Retro-AAV-DIO-eYFP show in **Figure 2M** displayed from a perspective of ventral to dorsal showing cells labeled in SuM and processes extending widely including in the hippocampus and POA. (100 and 500 μ m scale bars)



Supplemental Figure 3.

Combinatorial viral and genetic approach is effective with minimal background.

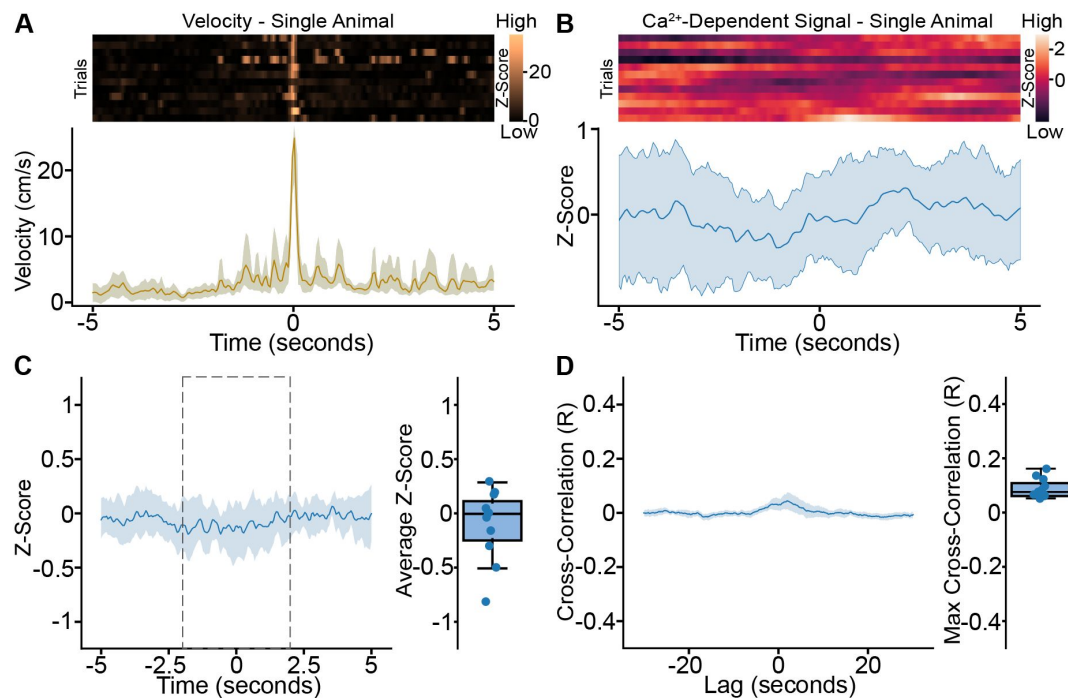
(A) Schematic of injection of Retro-AAV-Cre into the POA and AAV-Nuc-flox(mCherry)-eGFP into the SuM of WT mice. (B) In SuM eGFP (green) Cre positive and mCherry (red) Cre negative labeled neurons are seen based on expression of Cre mediated by POA injected retrograde AAV constructs showing SuM::POA neurons are a subset of all cells in SuM. (C and D) In wildtype mice injected in POA with Retro-AAV-fDIO-Cre and in SuM with AAV-DIO-ChR2eYFP expression of ChR2eYFP (green) is not seen. (E-H) In Ai14 reporter mice injected with AAV-fDIO-Cre into SuM tdTomato expression is not observed unless Retro-AAV2-Flp is injected. (100 μ m scale bars)



Supplemental Figure 4.

SuMVGLUT2⁺::POA neurons show Fos induction after FST and dunk test evokes active coping.

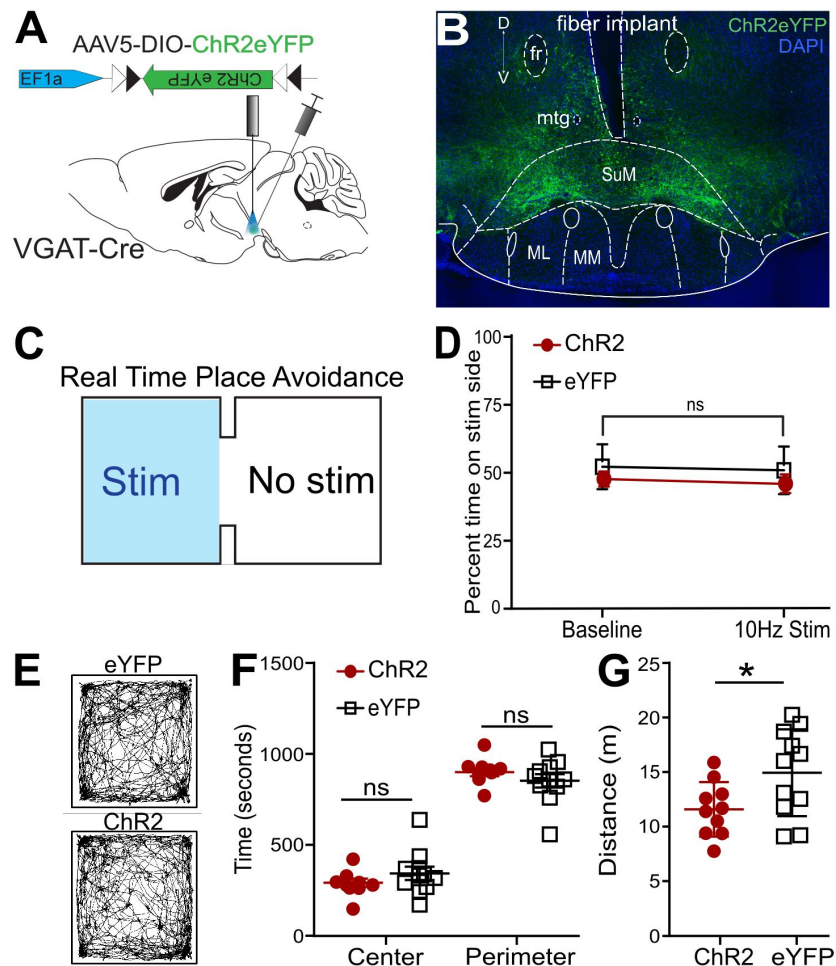
Sections of brains from mice (A) left in the home cage or (B) subjected to 15 minutes of forced swimming 90 min prior to sacrifice were probed with anti-Fos antibodies (cyan) showed (scale bars 500µm) (C) a significant ($p = 0.029$) increase in the mean $\pm$ SEM number of anti-Fos labeled neurons in brains from mice subjected to force swim ($n=5$) compared to control mice ($n=4$). (D) In VGLUT2-Cre mice Retro-AAV-DIO-mCherry was injected in the POA of mice ($n=5$) subjected to 15 minutes forced swim. (E) Region of SuM shown with higher magnification in (F) where SuM^{VGLUT2⁺::POA} neurons labeled by mCherry show overlap with cells (arrows) labeled by anti-Fos. (Scale bar 100µm) (G) A significant increase ($p = 0.016$) in the number of VGLUT2⁺ cells labeled with mCherry and anti-Fos in mice subjected to forced swim. (H) Mice subjected to 10 30-second trials of "dunk" forced swimming. (I) Quantitation of average time mobile for every 30-second trial during "dunk" forced swim.



Supplemental Figure 5.

SuMVG^{GLUT2+}::POA neurons are not recruited during increased locomotor activity in the absence of acute stressor.

(A) Heat maps showing velocity for 12 episodes of rapid locomotion from a representative animal during free movement in open field with mean ±95% CI for the trials. (B) For that same animal heat map of Ca²⁺ dependent signal was analyzed during average peak velocity showing no change in activity during peak velocity. Representative heat map for a single animal with the mean ±95% confidence interval of the 12 trials for same animal. (C) Mean ±95% confidence interval for Z-score of 12 trials for all 13 animals and Boxplot for average Z-score. (D) Cross-correlation analysis shows no correlation between Ca²⁺ signal from SuM^{VGLUT2+}::POA neurons spontaneous velocity increases.



Supplemental Figure 6.

GABAergic neurons in SuM do not drive place preference.

(A) Schematic of viral injection into VGAT-Cre mice in SuM. (B) VGAT+ neurons labeled and outline of implanted fiber. (C) Real-time arena showing the pair of photostimulation on one side of the arena. (D) Photostimulation at 10Hz did not drive a significant place preference. (eYFP (control) $n=26$ and ChR2=26). All data plotted as mean $\pm$ SEM. (E) Representative path traces of VGAT-Cre mice with ChR2 or eYFP (control) in open field testing. (F) No significant differences in center or perimeter time when comparing the ChR2eYFP expressing to controls VGAT-Cre mice. (G) Chr2eYFP expressing mice traveled slightly but significantly ($p=0.034$) less distance compared to control.

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Reviewer #1 (Public Review):

Summary:

This important manuscript investigates a subpopulation of glutamatergic neurons in the supramammillary nucleus that projects to the pre-optic hypothalamus area (SuM-VGLUT2+::POA). First, they define the neural circuitry of these neurons, which make contact with many stress/threat-associated brain regions. Then they employ fibre photometry to measure the activity of these neurons during various threatening tasks and find the responses correlate well with threat stimuli. Finally, they stimulate these neurons and find multiple lines of evidence that mice find this aversive and will act to avoid receiving this stimulation. In sum, they provide compelling evidence that this neuronal population represents a new node in stress response circuitry that allows the animal to produce flexible behaviours in response to stress, which will be of interest to neuroscientists across several sub-fields.

Strengths:

Overall I found a lot to like about this manuscript and very little to dislike. It is very novel and interesting, and the evidence given to support the conclusions is compelling.

Specific strengths:

- The topic is highly novel.
- The manuscript follows a logical structure and neatly moves through the central story. I found myself quite convinced of the evidence given for the conclusions that were made and many potential alternate interpretations are well-controlled for.
- The manuscript employs an array of different tasks to provide converging evidence for their conclusions.
- The authors provide excellent evidence of the specificity of the function of this neuronal

population, both from anatomical studies and from behavioural studies (e.g. demonstrating that activity of gabaergic neurons in the same region does not correlate with behaviours in the same way).

- The study is well-powered (sample sizes are good) and the effects are convincing.

Weaknesses:

- Despite the manuscript being generally well-written and easy to follow, there are several grammatical errors throughout that need to be addressed.
- Only p values are given in the text to support statistical differences. This is not sufficient. F and/or t values should be given as well. Moreover, the fibre photometry data does not appear to have any statistical analyses reported - only confidence intervals represented in the figures without any mention of whether the null hypothesis that the elevations in activity observed are different from the baseline. This is particularly important where there is ambiguity, such as in Figure 3K, where the spontaneous activity of the animal appears to correlate with a spike in activity but the text mentions that there is no such difference. Without statistics, this is difficult to judge.
- The use of photostimulation only is unfortunate, it would have been really nice to see some inactivation of these neurons as well. This is because of the well-documented issues with being able to determine whether photostimulation is occurring in a physiological manner, and therefore makes certain data difficult to interpret. For instance, with regards to the 'active coping' behaviours - is this really the correct characterisation of what's going on? I wonder if the mice simply had developed immobile responding as a coping strategy but when they experience stimulation of these neurons that they find aversive, immobility is not sufficient to deal with the summative effects of the aversion from the swimming task as well as from the neuronal activation? An inactivation study would be more convincing.
- Nose poke is only nominally instrumental as it cannot be shown to have a unique relationship with the outcome that is independent of the stimuli-outcome relationships (in the same way that a lever press can, for example). Moreover, there is nothing here to show that the behaviours are goal-directed.

Reviewer #2 (Public Review):

The manuscript by Escobedo et al. is an interesting investigation addressing the involvement of a lesser-studied brain region/neuron population (SUM glutamate neurons that project to the POA and other places) in active coping and locomotor behavior. The authors present data that this small population of glutamate neurons is an important circuit hub recruited for active coping but not overall locomotion by employing several behavioral tests. The manuscript is straightforward and potentially interesting, but the strength of the evidence and the significance of the paper as a whole is limited due to some lack of rigor with regards to 1) validation and quantification of anatomical tracing data that serve as a basis for the behavioral testing, 2) the use of statistics, 3) sex as a biological variable, 4) genotype differences between experimental and control groups in behavioral tests, and other concerns laid out below.

1. These are very difficult, small brain regions to hit, and it is commendable to take on the circuit under investigation here. However, there is no evidence throughout the manuscript that the authors are reliably hitting the targets and the spread is comparable across experiments, groups, etc., decreasing the significance of the current findings. There are no hit/virus spread maps presented for any data, and the representative images are cropped to avoid showing the brain regions lateral and dorsal to the target regions. In images where you can see the adjacent regions, there appears expression of cell bodies (such as Supp 6B), suggesting a lack of SuM specificity to the injections.

2. In addition, the whole brain tracing is very valuable, but there is very little quantification of the tracing. As the tracing is the first several figures and supp figure and the basis for the interpretation of the behavior results, it is important to understand things including how robust the POA projection is compared to the collateral regions, etc. Just a rep image for each of the first two figures is insufficient, especially given the above issue raised. the combination of validation of the restricted expression of viruses, rep images, and quantified tracing would add rigor that made the behavioral effects have more significance.

For example, in Fig 2, how can one be sure that the nature of the difference between the nonspecific anterograde glutamate neuron tracing and the Sum-POA glutamate neuron tracing is real when there is no quantification or validation of the hits and expression, nor any quantification showing the effects replicate across mice? It could be due to many factors, such as the spread up the tract of the injection in the nonspecific experiment resulting in the labeling of additional regions, etc.

Relatedly, in Supp 4, why isn't C normalized to DAPI, which they show, or area? Similar for G - what is the mcherry coverage/expression, and why isn't Fos normalized to that?

1. The authors state that they use male and female mice, but they do not describe the n's for each experiment or address sex as a biological variable in the design here. As there are baseline sex differences in locomotion, stress responses, etc., these could easily factor into behavioral effects observed here.
2. In a similar vein as the above, the authors appear to use mice of different genotypes (however the exact genotypes and breeding strategy are not described) for their circuit manipulation studies without first validating that baseline behavioral expression, habituation, stress responses are not different. Therefore, it is unclear how to interpret the behavioral effects of circuit manipulation. For example in 7H, what would the VGLUT2-Cre mouse with control virus look like over time? Time is a confound for these behaviors, as mice often habituate to the task, and this varies from genotype to genotype. In Fig 8H, it looks like there may be some baseline differences between genotypes- what is normal food consumption like in these mice compared to each other? Do Cre⁺ mice just locomote and/or eat less? This issue exists across the figures and is related to issues of statistics, potential genotype differences, and other experimental design issues as described, as well as the question about the possibility of a general locomotor difference (vs only stress-induced). In addition, the authors use a control virus for the control groups in VGAT-Cre manipulation studies but do not explain the reasoning for the difference in approach.
3. The statistics used throughout are inappropriate. The authors use serial Mann-Whitney U tests without a description of data distributions within and across groups. Further, they do not use any overall F tests even though most of the data are presented with more than two bars on the same graph. Stats should be employed according to how the data are presented together on a graph. For example, stats for pre-stim, stim, and post-stim behavior X between Cre⁺ and Cre⁻ groups should employ something like a two-way repeated measures ANOVA, with post-hoc comparisons following up on those effects and interactions. There are many instances in which one group changes over time or there could be overall main effects of genotype. Not only is serially using Mann-Whitney tests within the same panel misleading and statistically inaccurate, but it cherry-picks the comparisons to be made to avoid more complex results. It is difficult to comprehend the effects of the manipulations presented without more careful consideration of the appropriate options for statistical analysis.

Conceptual:

1. What does the signal look like at the terminals in the POA? Any suggestion from the data that the projection to the POA is important?
2. Is this distinguishing active coping behavior without a locomotor phenotype? For example, Fig. 5I and other figure panels show a distance effect of stimulation (but see issues raised about the genotype of comparison groups). In addition, locomotor behavior is not included for many behaviors, so it is hard to completely buy the interpretation presented.
3. What is the role of GABA neurons in the SuM and how does this relate to their function and interaction with glutamate neurons? In Supp 8, GABA neuron activation also modulates locomotion and in Fig 7 there is an effect on immobility, so this seems pretty important for the overall interpretation and should probably be mentioned in the abstract.

Questions about figure presentation:

1. In Fig 3, why are heat maps shown as a single animal for the first couple and a group average for the others? Why is the temporal resolution for J and K different even though the time scale shown is the same? What is the evidence that these signal changes are not due to movement per se?
2. In Fig 4, the authors carefully code various behaviors in mice. While they pick a few and show them as bars, they do not show the distribution of behaviors in Cre- vs Cre+ mice before manipulation (to show they have similar behaviors) or how these behaviors shift categories in each group with stimulation. Which behaviors in each group are shifting to others across the stim and post-stim periods compared to pre-stim?
Of note, issues of statistics, genotype, and SABV are important here. For example, the hint that treading/digging may have a slightly different pre-stim basal expression, it seems important to first evaluate strain and sex differences before interpreting these data.
3. Why do the authors use 10 Hz stimulation primarily? is this a physiologically relevant stim frequency? They show that they get effects with 1 Hz, which can be quite different in terms of plasticity compared to 10 Hz.
4. In Fig 5A-F, it is unclear whether locomotion differences are playing a role. Entrances (which are low for both groups) are shown but distance traveled or velocity are not.

In B, there is no color in the lower left panel. where are these mice spending their time? How is the entirety of the upper left panel brighter than the lower left? If the heat map is based on time distribution during the session, there should be more color in between blue and red in the lower left when you start to lose the red hot spots in the upper left, for example. That is, the mice have to be somewhere in apparatus. If the heat map is based on distance, it would seem the Cre- mice move less during the stim.

1. By starting with 1 hz, are the experimenters inducing LTD in the circuit? what would happen if you stop stimming after the first epoch? Would the behavioral effect continue? What does the heat map for the 1 hz stim look like?

Relatedly, it is a lot of consistent stimulation over time and you likely would get glutamate depletion without a break in the stim for that long.

1. In Fig 6, the authors show that the Cre- mice just don't do the task, so it is unclear what the utility of the rest of the figure is (such as the PR part). Relatedly, the pause is dependent on the activation, so isn't C just the same as D? In G and H, why is a subset of Cre+ mice shown? Why not all mice, including Cre- mice?
2. In Fig 7, what does the GCaMP signal look like if aligned to the onset of immobility? It looks like since the hindpaw swimming is short and seems to precede immobility, and the increase in the signal is ramping up at the onset of hindpaw swimming, it may be that the calcium signal is aligned with the onset of immobility. What does it look like for swimming onset? In I, what is the temporal resolution for the decrease in immobility? Does it start prior to the termination of the stim, or does it require some elapsed time after the termination, etc?

Reviewer #3 (Public Review):

Summary:

Coping with stress by the animal in danger is essential for survival. The current study identified a novel population of neurons in the murine supramammillary nucleus (SuM) projecting to the POA as well as diverse brain regions relevant to the decision-making by combinatorial labeling of the neurons with adeno-associated viruses (AAVs). Such a unique population of glutamatergic neurons was activated under a variety of acute stress, while the optogenetic stimulation of them induced behaviors relevant to the active coping of the stress.

Strengths:

Discovery of the neural circuit converting the passive to the active stress coping strategy of the behavior in this study will provide deep insight into understanding how the animal survives with flexibility and must be informative for the neuroscience community.

Weaknesses:

Despite a large advance in understanding the role of this circuit in behavior in the study, I primarily have concerns about the interaction between SuM and other neural pathways.